

Electronics Device & Circuits

EE 223

Reference material for Digital Electronics

by

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Outcome 1: To understand basics of Digital logic devices

- Electronics circuits are broadly classified into analog and digital
- Analog electronics process signals with continuous values
- Digital electronics process signals with discrete values
- Most things that can be measured quantitatively appear in nature in analog form
- Example : air temperature, voltage signal
- Draw sine wave in analog form and digital form (20ms)
- Each 2ms take a sample, where each sample is represented by a digital code

Outcome 1: To understand basics of Digital logic devices

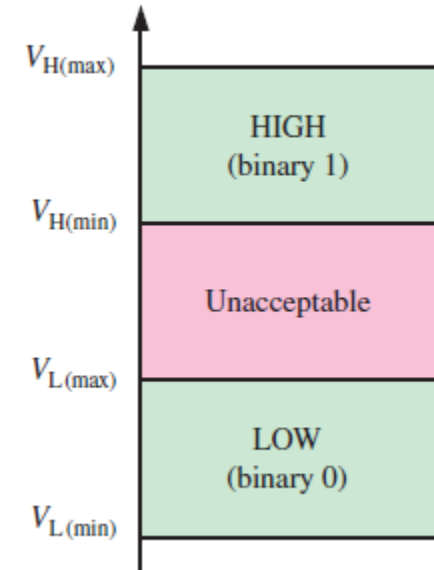
- Advantage of digital representation over analog representation in electronics applications
 - Digital data can be processed and transmitted more efficiently and reliably
 - Digital data can be store compactly
 - Digital data can be reproduced with greater accuracy
 - Digital data are least affect by noise or unwanted voltage fluctuations
- Digital electronics contains only two possible states:
 - HIGH
 - LOW
- Boolean algebra is used for design and analysis of digital systems.

Outcome 1: To understand basics of Digital logic devices

- Digital electronics is represented by **binary number system**
- 0 and 1 forms two digits of the binary number system
- A binary digit is called as a bit. Groups of bits are called codes.
- Codes are used to represent a number, letter, symbol, instructions in any given applications.
- In digital circuits, two different voltage levels are used to represent two bits.
- 1 represent HIGH and 0 represent LOW.

Outcome 1: To understand basics of Digital logic devices

- In a practical digital circuit, HIGH is a value between $V_H(\text{max})$ and $V_H(\text{min})$. Similarly, LOW is a value between $V_L(\text{max})$ and $V_L(\text{min})$.
- For example if 5V is HIGH and 0V is LOW for an digital IC. Voltages in the range of 4.2V to 5.2V and voltages from 0V to 0.8V may be represented as HIGH and LOW for the digital IC, These datas are generally mentioned in the data sheet of any IC.



Outcome 1: To understand basics of Digital logic devices

- Binary information handled by digital system appears as waveform.
- Digital waveform consist of voltage levels changing back and forth between the HIGH and LOW levels or states.
- The pulses are ideal because the rising and falling edges are assumed to change in zero time (instantaneously).

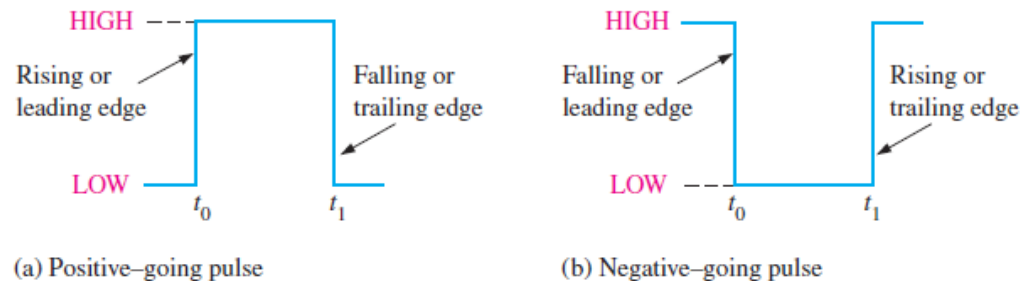


FIGURE 1-7 Ideal pulses.

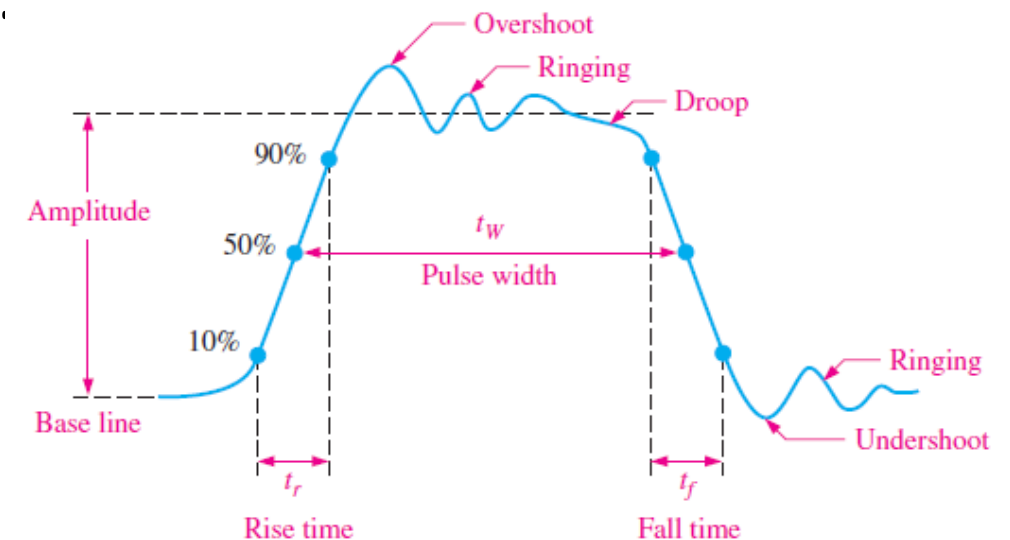


FIGURE 1-8 Nonideal pulse characteristics.

Outcome 1: To understand basics of Digital logic devices

- The digital waveform is composed of series of pulses called as pulse train.
- These pulse train can be periodic or nonperiodic. Draw in board
- Periodic pulse waveform repeat itself at a fixed interval called time period.
- Frequency is the rate at which the pulses repeat. Measured in Hz.
- Frequency and Time Period are inversely proportional.
- Another important characteristic of periodic waveform is duty cycle. Ratio of pulse width to time period.

Outcome 1: To understand basics of Digital logic devices

- In digital system, all waveforms are synchronized with clock.
- Clock is a period waveform as shown below.
- Each change in level of waveform A occurs at leading edge of clock waveform
- A group of several bits can be used as a piece of information such as number or a letter.

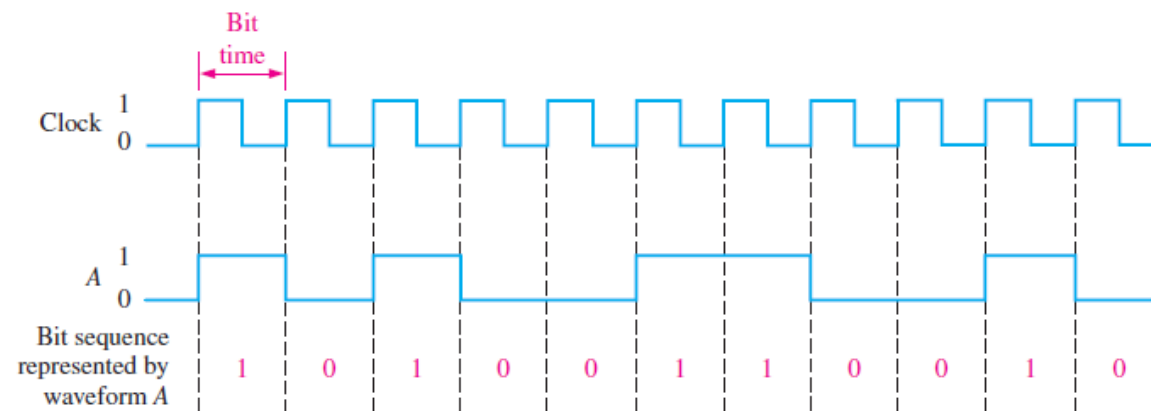


FIGURE 1-11 Example of a clock waveform synchronized with a waveform representation of a sequence of bits.

Outcome 1: To understand basics of Digital logic devices

- In Decimal number system, the position of each digit in a decimal number indicates the magnitude of the quantity represented and can be assigned a weight.
- The weight for whole numbers are positive powers of ten that increase from right to left, beginning with $10^0 = 1$.
- For fractional number, the weight are negative powers of ten that decrease from left to right beginning with 10^{-1} .
- The value of a decimal number is sum of digits after each digit has been multiplied by its weight. Example 47, 568.23

Outcome 1: To understand basics of Digital logic devices

- Binary number system is another way to represent quantities.
- Decimal system with ten digits is a base ten system. Binary system with two digits is a base two system.
- 3 bits are required to count from 0 to 8
- $2^n - 1$, where n is number of bits
- With 6 bits upto 63 decimal number can be counted

TABLE 2-1

Decimal Number	Binary Number			
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

Outcome 1: To understand basics of Digital logic devices

- Similar to weights in decimal system, the position of 1 and 0 indicates its weight.

$$2^{n-1} \dots 2^3 2^2 2^1 2^0 . 2^{-1} 2^{-2} \dots 2^{-n}$$


Binary point

The right-most bit is the **LSB** (least significant bit) in a binary whole number and has a weight of $2^0 = 1$. The weights increase from right to left by a power of two for each bit. The left-most bit is the **MSB** (most significant bit); its weight depends on the size of the binary number.

Binary-to-Decimal Conversion

The decimal value of any binary number can be found by adding the weights of all bits that are 1 and discarding the weights of all bits that are 0.

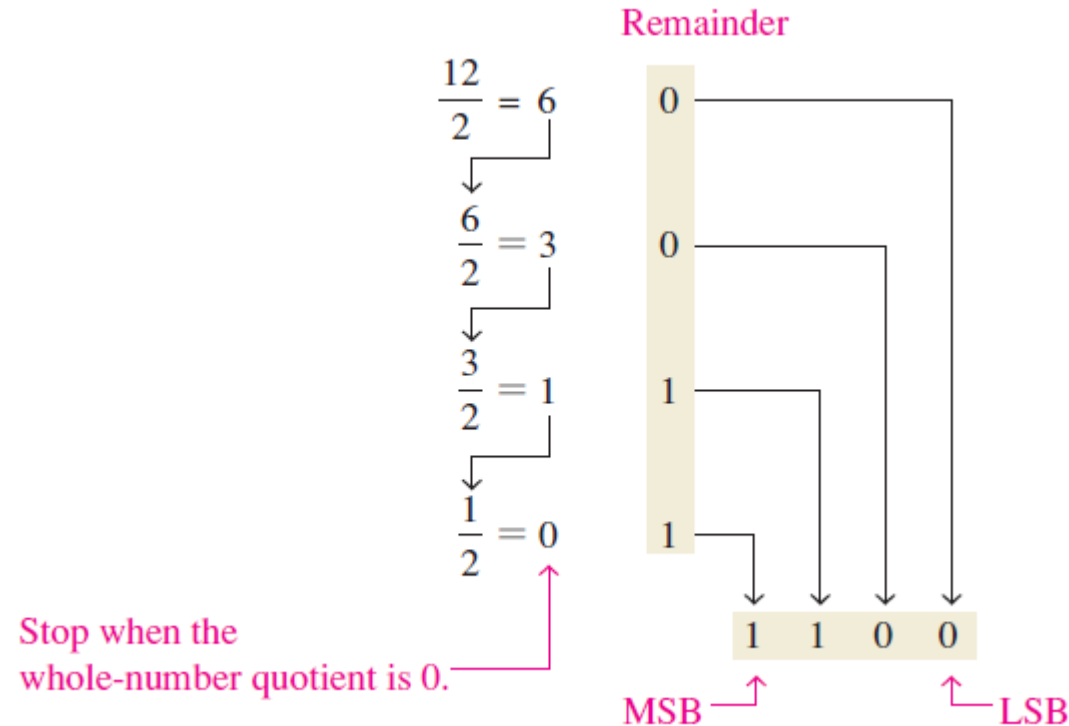
Example : 1101101 – 109

0.1011 – 0.6875

Outcome 1: To understand basics of Digital logic devices

- Decimal-to-Binary Conversion

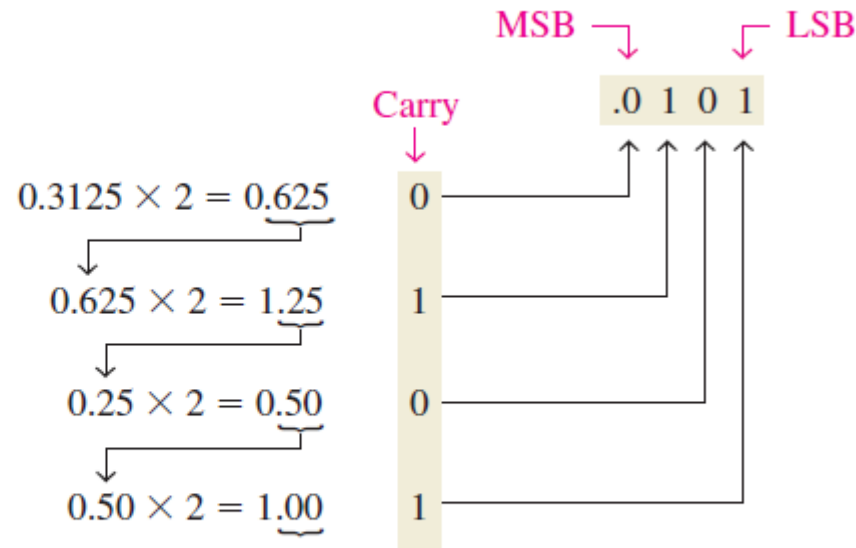
Example : 19 = 10011
45 = 101101



Outcome 1: To understand basics of Digital logic devices

- Converting Decimal Fractions to Binary

Example:
 $0.625 = 0.101$
 0.375



Continue to the desired number of decimal places or stop when the fractional part is all zeros.

Outcome 1: To understand basics of Digital logic devices

- Binary arithmetic contains
 - binary addition,
 - Binary subtraction,
 - Binary multiplication,
 - Binary division

Binary Addition

$0 + 0 = 0$	Sum of 0 with a carry of 0
$0 + 1 = 1$	Sum of 1 with a carry of 0
$1 + 0 = 1$	Sum of 1 with a carry of 0
$1 + 1 = 10$	Sum of 0 with a carry of 1

Add the following binary numbers:

(a) $11 + 11$ **(b)** $100 + 10$

(c) $111 + 11$ **(d)** $110 + 100$

Outcome 1: To understand basics of Digital logic devices

Binary Subtraction

$0 - 0 = 0$
 $1 - 1 = 0$
 $1 - 0 = 1$
 $10 - 1 = 1$ $0 - 1$ with a borrow of 1

Subtract 011 from 101.
101 from 110.

(a) $11 - 01$

(b) $11 - 10$

Solution

(a)
$$\begin{array}{r} 11 \\ -01 \\ \hline 10 \end{array}$$

(b)
$$\begin{array}{r} 11 \\ -10 \\ \hline 01 \end{array}$$

Let's examine exactly what was done to subtract the two binary numbers since a borrow is required. Begin with the right column.

Left column:

When a 1 is borrowed, a 0 is left, so $0 - 0 = 0$.

Middle column:

Borrow 1 from next column to the left, making a 10 in this column, then $10 - 1 = 1$.

Right column:

$1 - 1 = 0$

$$\begin{array}{r} 0101 \\ -011 \\ \hline 010 \end{array}$$

Outcome 1: To understand basics of Digital logic devices

Binary Multiplication

$$\begin{array}{l} 0 \times 0 = 0 \\ 0 \times 1 = 0 \\ 1 \times 0 = 0 \\ 1 \times 1 = 1 \end{array}$$

Multiply $11 * 11$
 $101 * 111$

(a) 11×11

(b) 101×111

Solution

(a)

$$\begin{array}{r} 11 \\ \times 11 \\ \hline 11 \\ +11 \\ \hline 1001 \end{array}$$

Partial products

$$\begin{array}{r} 3 \\ \times 3 \\ \hline 9 \end{array}$$

(b)

$$\begin{array}{r} 111 \\ \times 101 \\ \hline 111 \\ 000 \\ +111 \\ \hline 10011 \end{array}$$

Partial products

Related Problem

Multiply 1101×1010 .

Outcome 1: To understand basics of Digital logic devices

Binary Division

Division in binary follows the same procedure as division in decimal

Divide 110/11

Solution

	10	2		11	3
(a)	11 $\overline{)110}$	3 $\overline{)6}$	(b)	10 $\overline{)110}$	2 $\overline{)6}$
	11	6		10	6
	<u>000</u>	<u>0</u>		<u>10</u>	<u>0</u>
				10	
				<u>00</u>	

Related Problem

Divide 1100 by 100.

Outcome 1: To understand basics of Digital logic devices

- Digital system handle both positive and negative number
- Signed binary number contains sign and magnitude information
- Sign indicates if the binary number is +ve or -ve
- Magnitude represents value of the number
- 3 forms to represent negative numbers in binary
 - Sign magnitude form
 - 1s complement
 - 2s complement
- MSB indicates the sign: 0 is +ve and 1 is -ve

Outcome 1: To understand basics of Digital logic devices

- **Sign Magnitude form**
- When a signed binary number is represented in sign-magnitude, the left-most bit is the sign bit and the remaining bits are the magnitude bits.
- +25 is represented as 00011001
- -25 is represented as 10011001
- **In the sign-magnitude form, a negative number has the same magnitude bits as the corresponding positive number but the sign bit is a 1 rather than a zero.**

Outcome 1: To understand basics of Digital logic devices

- 1s complement and 2s complement of a binary number
- Importance: Representation of negative numbers
- 2s complement is generally used in computers to handle negative numbers
- 1s complement can be found by interchanging all 1s to 0s and 0s to 1s.
- Example : 00011001 (+25)
- 2s complement = 1s complement + 1
- 2s complement alternate method : From LSB towards MSB write the bits as they are upto first 1. Then perform 1s complement on remaining bits upto MSB
- Problem (0010 1101)(+45)

Outcome 1: To understand basics of Digital logic devices

- The Decimal Value of Signed Numbers
- **Sign-Magnitude**
- by summing the weights in all the magnitude bit positions where there are 1s and ignoring those positions where there are zeros.
- Example: 10010101 (-21)
- **1s Complement**
- by assigning a negative value to the weight of the sign bit, summing all the weights where there are 1s, and adding 1 to the result.

Outcome 1: To understand basics of Digital logic devices

Determine the decimal values of the signed binary numbers expressed in 1's complement:

- (a) 00010111 (b) 11101000

Solution

- (a) The bits and their powers-of-two weights for the positive number are as follows:

-2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
0	0	0	1	0	1	1	1

Summing the weights where there are 1s,

$$16 + 4 + 2 + 1 = +23$$

- (b) The bits and their powers-of-two weights for the negative number are as follows.
Notice that the negative sign bit has a weight of -2^7 or -128 .

-2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1	1	1	0	1	0	0	0

Summing the weights where there are 1s,

$$-128 + 64 + 32 + 8 = -24$$

Adding 1 to the result, the final decimal number is

$$-24 + 1 = -23$$

Outcome 1: To understand basics of Digital logic devices

- **2s complement**
- are determined by summing the weights in all bit positions where there are 1s and ignoring those positions where there are zeros.

EXAMPLE 2-17

Determine the decimal values of the signed binary numbers expressed in 2's complement:

- (a) 01010110 (b) 10101010

Solution

- (a) The bits and their powers-of-two weights for the positive number are as follows:

-2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
0	1	0	1	0	1	1	0

Summing the weights where there are 1s,

$$64 + 16 + 4 + 2 = +86$$

- (b) The bits and their powers-of-two weights for the negative number are as follows. Notice that the negative sign bit has a weight of $-2^7 = -128$.

-2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1	0	1	0	1	0	1	0

Summing the weights where there are 1s,

$$-128 + 32 + 8 + 2 = -86$$

Related Problem

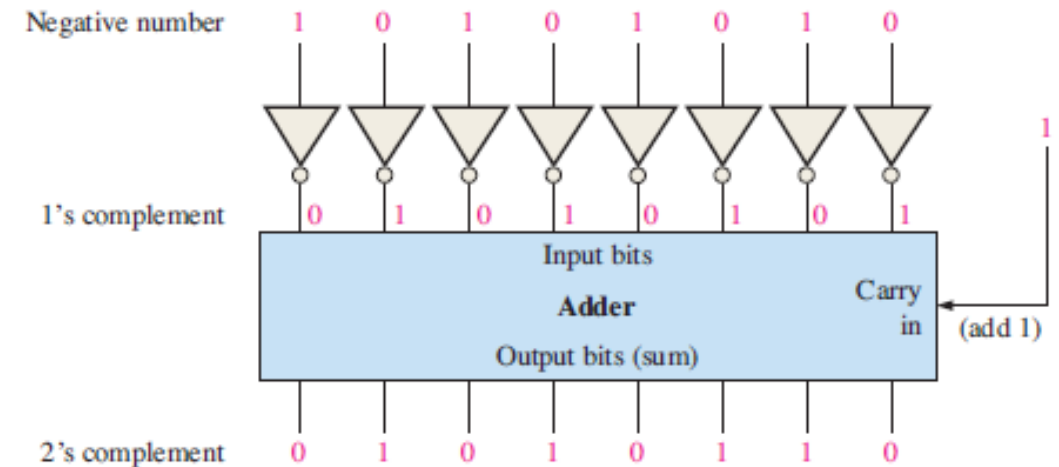
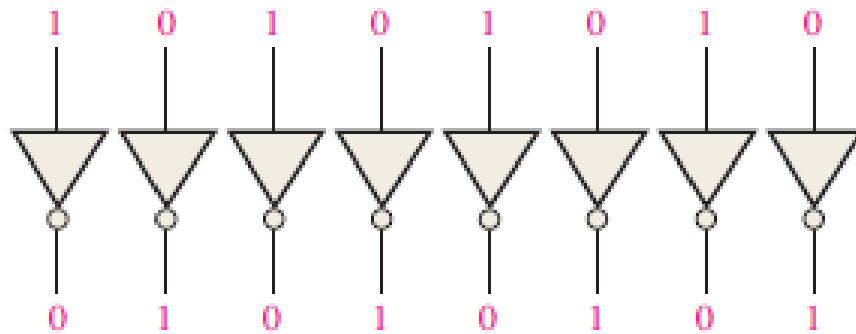
Determine the decimal value of the 2's complement number 11010111.

Outcome 1: To understand basics of Digital logic devices

- Advantage of 2's complement
- it simply requires a summation of weights regardless of whether the number is positive or negative.
- the 1's complement form is generally not used because two representations of zero (00000000 or 11111111) are possible.

Outcome 2

- Realization of 1s complement and 2s complement for 8 bit number



To return back to the original number just perform same procedure on the resultant binary number

Outcome 1: To understand basics of Digital logic devices

- Arithmetic operations with signed numbers
- Addition involves adding two numbers and discarding final carry bit
 - Both numbers are positive
 - $(00000111 + 00000100) = 00001011 \Rightarrow 7 + 4 = 11$
 - Positive number with magnitude greater than negative number
 - $00001111 + 11111010 = 00001001 \Rightarrow 15 - 6 = 9$
 - Discard the carry bit
 - Positive number with magnitude less than negative number
 - $00010000 + 11101000 = 11111000 \Rightarrow 16 - 24 = -8$
 - Discard the carry bit and answer is in 2's complement form
 - Both negative numbers
 - $11111011 + 11110111 = 11110010 \Rightarrow -5 - 9 = -14$
 - Discard the carry bit and answer is in 2's complement form

Outcome 1: To understand basics of Digital logic devices

- Overflow condition:
- The condition when the resultant number of bits exceeds the provided number of bits, while adding two numbers, is known as overflow condition.
- Consider a case, where the system works with 8 bit numbers only
 - MSB as sign bit and other 7 bits as magnitude bit
- $01111101 + 00111010 = 10110111 \Rightarrow 125 + 58 = 183$
- 183 is represented using 8 magnitude bits.
- Carry into sign bit indicates overflow condition

Outcome 1: To understand basics of Digital logic devices

- Subtraction of signed binary number
- Special case of addition
- Perform 2's complement on the negative number and do addition
- $00001000 - 00000011 = 00000101 \Rightarrow 8 + (-3) = 5$
- $10001000 - 11100010 = 10100110 \Rightarrow -120 - (-30) = -90$
- Discard the final carry bit in the resultant

Outcome 1: To understand basics of Digital logic devices

- Multiplication of signed binary numbers
- Also performed using addition
- Two methods: direct addition and partial product
- Direct addition
 - $00001000 \times 00000011 = 00011000$ (8 (multiplicand) $\times$ 3 (multiplier)) = $8+8+8$
 - Disadvantage: very complex in case of large number multiplier (3)
- Partial Product
 - The multiplicand is multiplied by each multiplier digit starting from LSD
 - Forming partial products
 - Each partial product is shifted one place left and then addition is performed

239	Multiplicand
<u>$\times 123$</u>	Multiplier
717	1st partial product (3×239)
478	2nd partial product (2×239)
<u>$+ 239$</u>	3rd partial product (1×239)
29,397	Final product

Outcome 1: To understand basics of Digital logic devices

- Partial Product:
- Step 1: Check the sign of multiplicand and multiplier
 - If both multiplicand and multiplier have same sign, then the resultant is positive
 - If both multiplicand and multiplier have different sign, then the product is negative
- Step 2: Convert the negative number if any to 2's complement
- Step 3: Generate partial products.
 - When the multiplier bit is 1, the partial product is the same as the multiplicand.
 - When the multiplier bit is 0, the partial product is zero.
 - Shift each successive partial product one bit to the left.
- Step 4: Add all the partial product
- Step 5: In step 1, if resultant is negative, perform 2's complement on the resultant and attach sign bit. If resultant is +ve, leave as such

Outcome 1: To understand basics of Digital logic devices

- Partial Product: $0011 \times 1110 \Rightarrow (3 \times -2)$
- Step 1: Resultant is –ve number
- Step 2: 2's complement of 1110 = 0010
- Step 3: $P1 = 0000$, $P2 = 0011$, $P3 = 0000$, $P4 = 0000$
- Step 4: $P1+P2+P3+P4 = 0000110$
- Step 5 : Perform 2's complement on step 4 : 1111010 and attach sign bit 11111010 (-6)
- 4 bit number multiplied by 4 bit number requires atleast 8 bits to store the resultant value

Perform 6×-8

Outcome 1: To understand basics of Digital logic devices

- Division of signed binary numbers
- Division can be performed using subtraction, and subtraction can be performed using addition
- Consider 00010101 divided by 11111001 $\Rightarrow 21/-7$
- Step 1: Check sign of both dividend and divisor, resultant –ve.
- Step 2: Perform 2's complement on –ve number
- Step 3: Initialize quotient to zero
- Step 4: Subtract the divisor from the dividend using 2's complement addition to get the first partial remainder and add 1 to the quotient. If the partial remainder is zero or negative, the division is complete. Else continue step 4
- Step 5: Perform 2's complement on resultant

After Mini Test

Outcome 1: To understand basics of Digital logic devices

- Hexadecimal number system
- 16 digit number system
- Compact representation of binary number system
 - long binary numbers are difficult to read and write
- Binary to hexadecimal conversion
 - Split binary number into 4-bit groups, starting at the LSB and replace each 4-bit group with the equivalent hexadecimal symbol.
- Hexadecimal to binary conversion
 - Replace hexadecimal number with 4 bit binary

TABLE 2-3		
Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Outcome 1: To understand basics of Digital logic devices

- Octal number system
- 8 digit number system
- Octal 2 binary conversion
- Replace each octal digit with the appropriate three bits binary.
- Binary 2 Octal conversion
 - Start from LSB, group three bits and, moving from right to left, convert each 3-bit group to the equivalent octal digit.
 - If there are not three bits available for the left-most group, add either one or two zeros to make a complete group

TABLE 2-4

Octal/binary conversion.

Octal Digit	0	1	2	3	4	5	6	7
Binary	000	001	010	011	100	101	110	111

Outcome 1: To understand basics of Digital logic devices

- Binary Coded Decimal (BCD)
- 4 bits of binary used to represent a decimal digit
- Sixteen numbers (0000 through 1111) can be represented.
- The six code combinations that are not used—1010, 1011, 1100, 1101, 1110, and 1111—are invalid in the BCD code.

TABLE 2-5

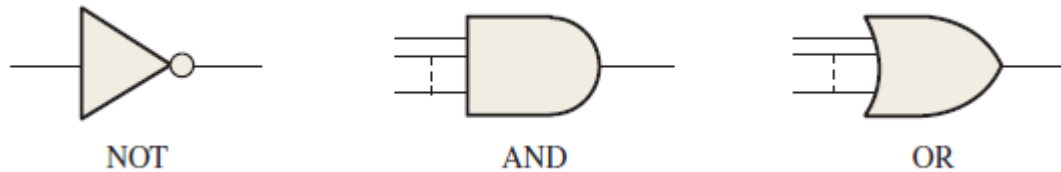
Decimal/BCD conversion.

Decimal Digit	0	1	2	3	4	5	6	7	8	9
BCD	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001

1001010001110000

Outcome 1: To understand basics of Digital logic devices

- Logic gates are one of the fundamental building blocks of digital systems.
- The three basic logic functions are NOT, AND and OR.

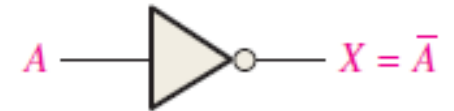
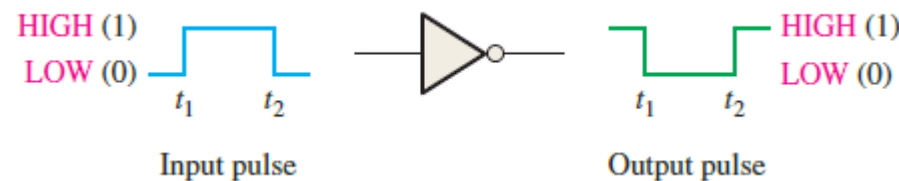


- A circuit that performs a logic function is known as logic gates
- Inverter also known as NOT gate is the logic circuit that performs NOT logic function
- AND and OR gate perform AND and OR logic function. These logic gates can have as many number of inputs

Outcome 1: To understand basics of Digital logic devices

- In any logic circuit, generally the inputs are on the left of the logic symbol and outputs are on the right
- Truth Table for NOT logic function states its operation

Input	Output
0	1
1	0

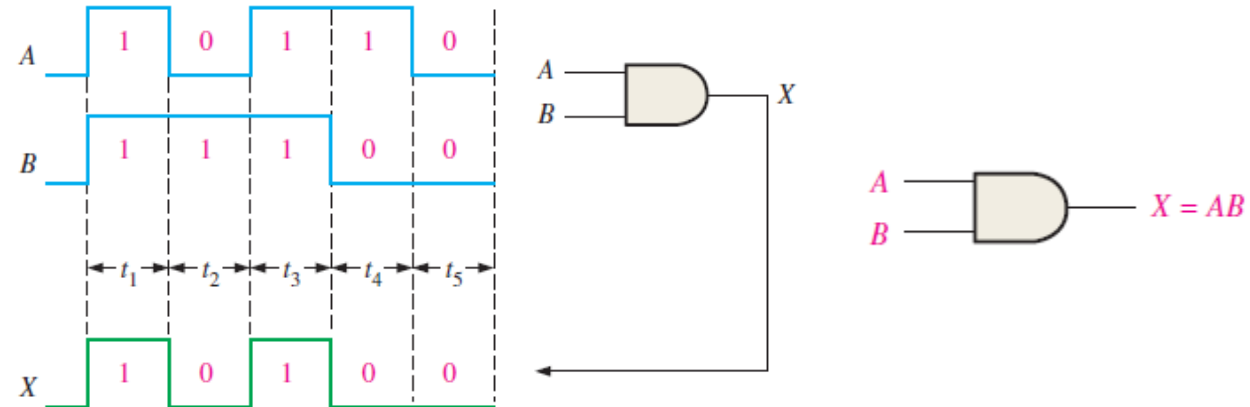


- Logic expression for Inverter is performed using Boolean algebra
- Boolean algebra is the mathematics of logic circuit
- A is the input and X is the output. Bar over a letter indicates complement function

Outcome 1: To understand basics of Digital logic devices

- AND Gate performs binary multiplication
- The logical operation of AND gate is indicated using truth table

Input 1	Input 2	Output
0	0	0
0	1	0
1	0	0
1	1	1

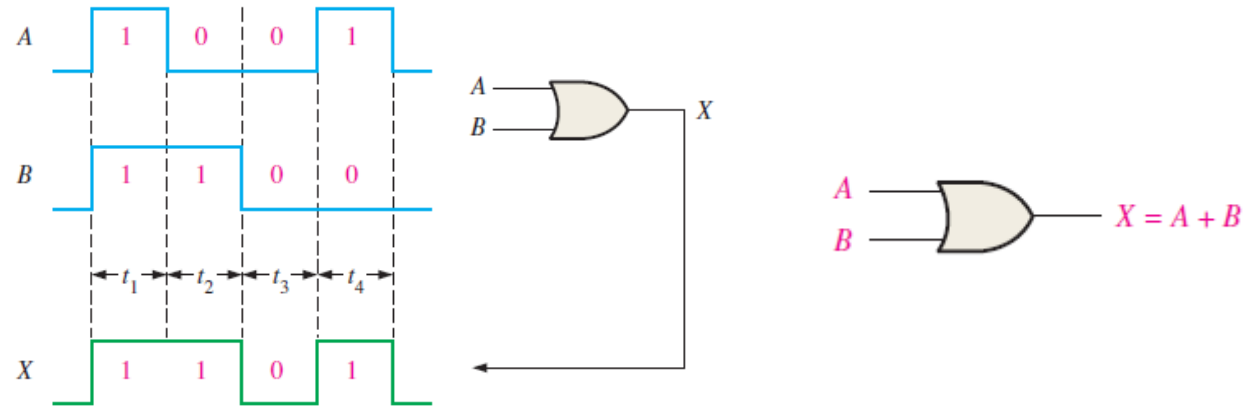


- AND gate can have any number of inputs (n). The total input combinations (N) is 2^n
- Mathematical representation is A.B or AB

Outcome 1: To understand basics of Digital logic devices

- OR gate performs logical addition
- The logical operation of OR gate is indicated using truth table

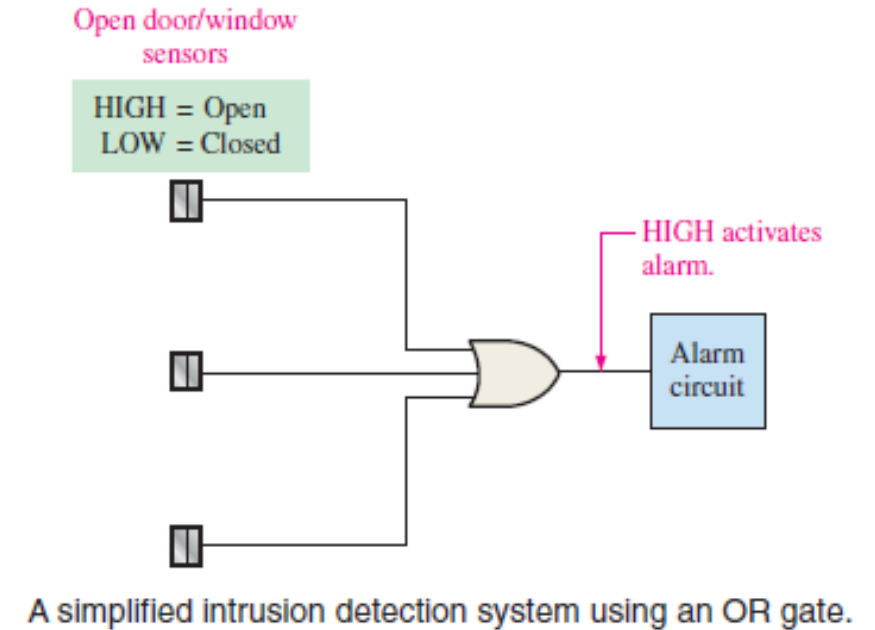
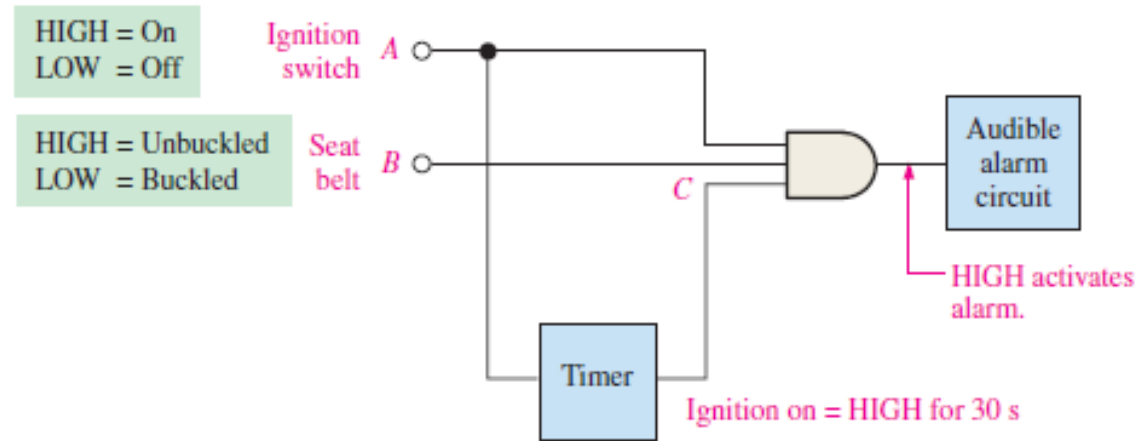
Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	1



- Mathematical representation is $A+B$

Outcome 2 & 4

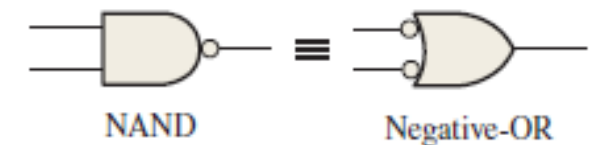
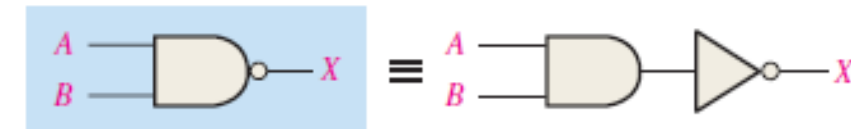
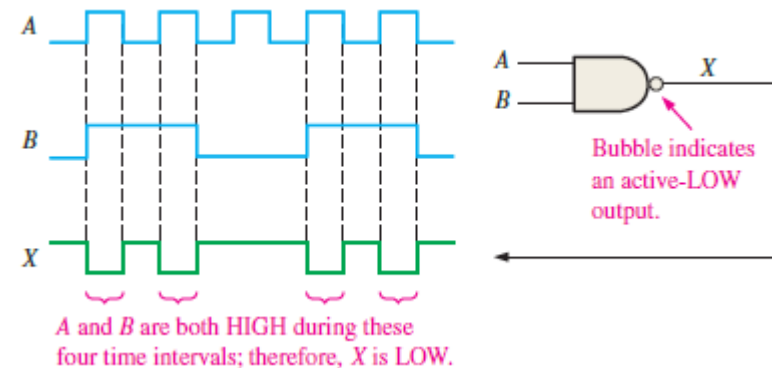
- Application of AND and OR Gate



Outcome 1: To understand basics of Digital logic devices

- NAND is an universal gate.
- The logical operation of NAND gate is indicated using truth table

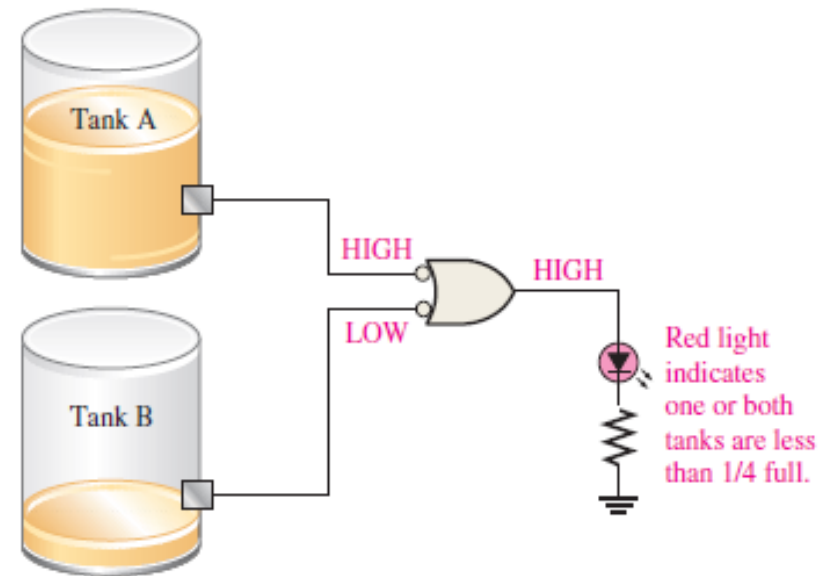
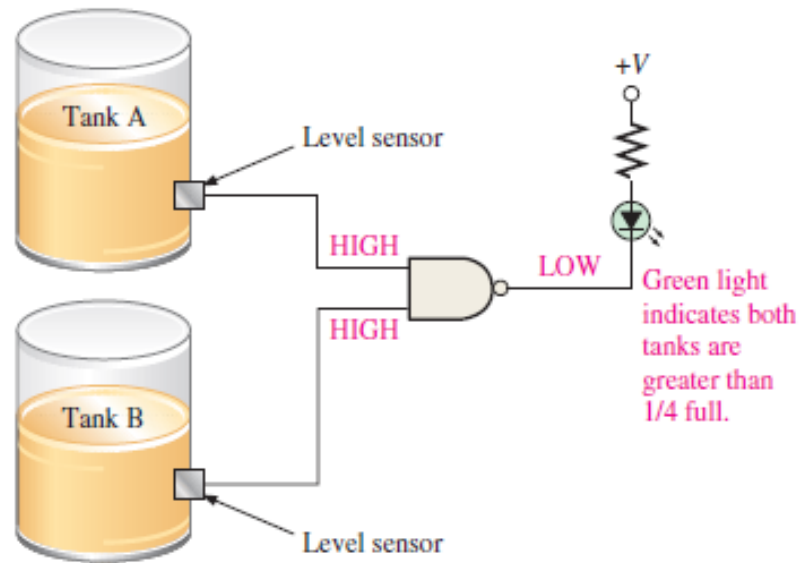
Input 1	Input 2	Output
0	0	1
0	1	1
1	0	1
1	1	0



- Mathematical representation is complement of (AB)
- It can also be termed as negative OR gate

Outcome 2

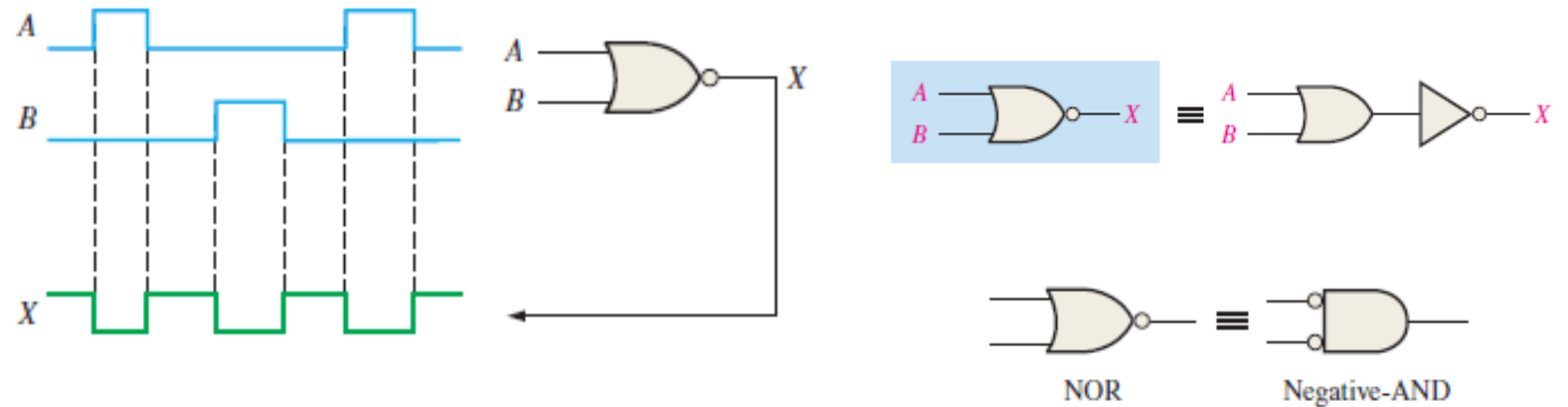
- Application of NAND Gate



Outcome 1: To understand basics of Digital logic devices

- NOR is an universal gate.
- The logical operation of NOR gate is indicated using truth table

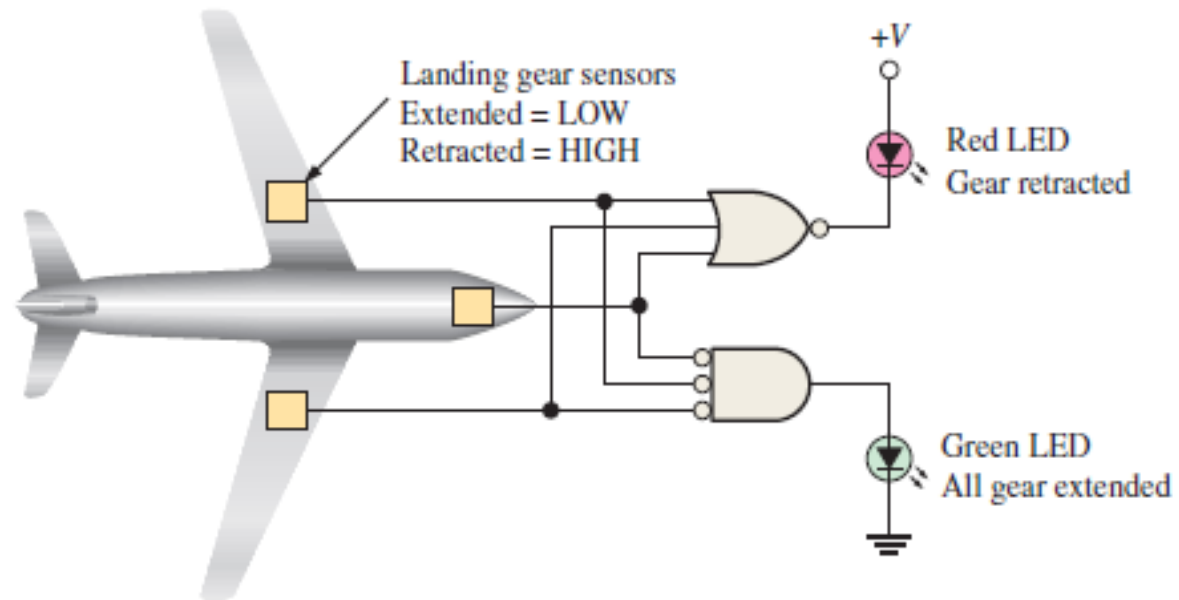
Input 1	Input 2	Output
0	0	1
0	1	0
1	0	0
1	1	0



- Mathematical representation is complement of $(A+B)$
- It can also be termed as negative AND gate

Outcome 2 & 4

- Application of NOR Gate



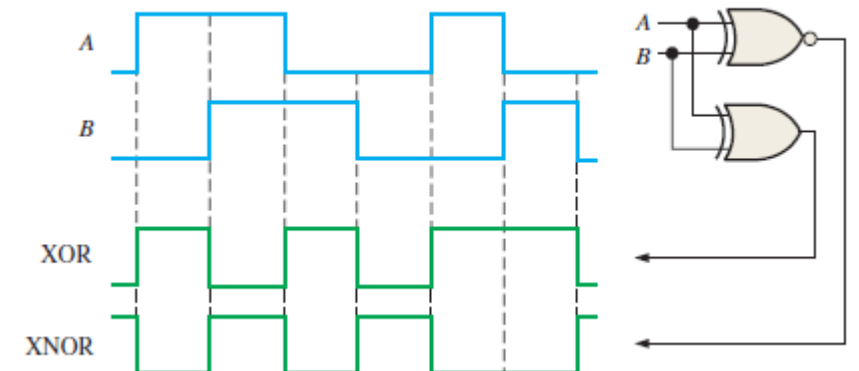
Outcome 1: To understand basics of Digital logic devices

- Exclusive-OR (XOR) and Exclusive-NOR (XNOR) are a two input gate.
- The logical operation of XOR and XNOR gate is indicated using truth table

Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	0

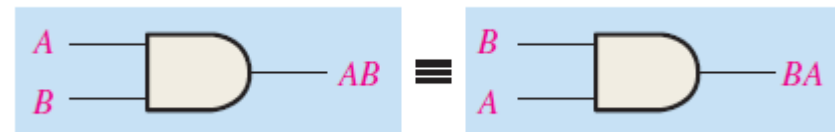
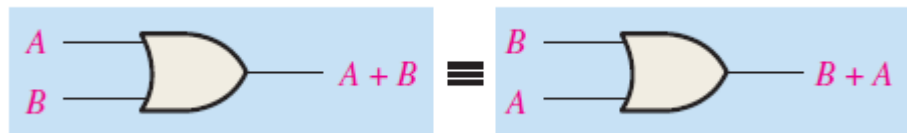


Input 1	Input 2	Output
0	0	1
0	1	0
1	0	0
1	1	1



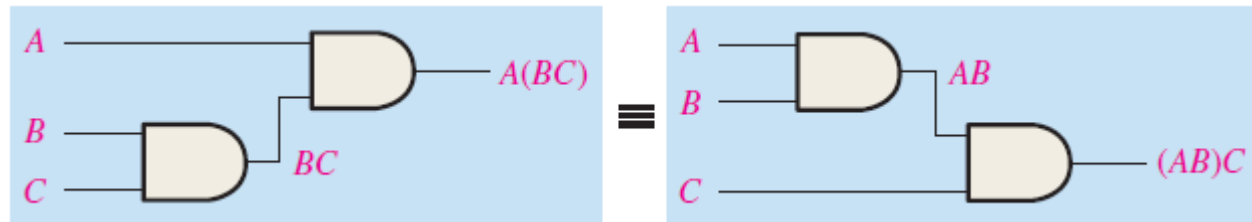
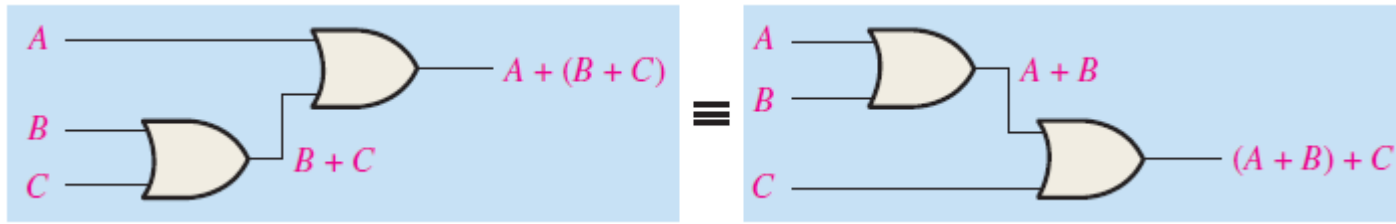
Outcome 1: To understand basics of Digital logic devices

- Boolean Algebra is the mathematics of digital logic
- Boolean addition is equivalent to the OR operation
- Boolean multiplication is equivalent to the AND operation
- Basic laws of Boolean algebra for addition and multiplication are
 - Commutative law
 - Associative law
 - Distributive law
- Commutative Law states that order in which the variables are ORed or ANDed makes no difference.



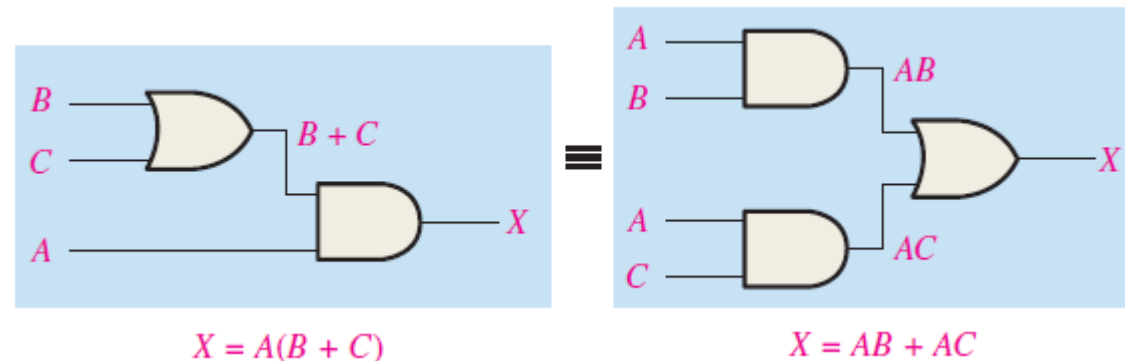
Outcome 1: To understand basics of Digital logic devices

- Associative law states that when ORing or ANDing more than two variables, the result is the same regardless of the grouping of the variables.



Outcome 1: To understand basics of Digital logic devices

- **Distributive Law** states that ORing two or more variables and then ANDing the result with a single variable is equivalent to ANDing the single variable with each of the two or more variables and then ORing the products.
- Also expresses the process of *factoring* in which the common variable A is factored out of the product terms



Outcome 1: To understand basics of Digital logic devices

- Rules of Boolean Algebra

Basic rules of Boolean algebra.

1. $A + 0 = A$

2. $A + 1 = 1$

3. $A \cdot 0 = 0$

4. $A \cdot 1 = A$

5. $A + A = A$

6. $A + \bar{A} = 1$

7. $A \cdot A = A$

8. $A \cdot \bar{A} = 0$

9. $\bar{\bar{A}} = A$

10. $A + AB = A$

11. $A + \bar{A}B = A + B$

12. $(A + B)(A + C) = A + BC$

Rule 11: $A + \bar{A}B = A + B$ This rule can be proved as follows:

$$A + \bar{A}B = (A + AB) + \bar{A}B$$

$$= (AA + AB) + \bar{A}B$$

$$= AA + AB + A\bar{A} + \bar{A}B$$

$$= (A + \bar{A})(A + B)$$

$$= 1 \cdot (A + B)$$

$$= A + B$$

Rule 10: $A = A + AB$

Rule 7: $A = AA$

Rule 8: adding $A\bar{A} = 0$

Factoring

Rule 6: $A + \bar{A} = 1$

Rule 4: drop the 1

Logic Simplification Using Boolean Algebra

Rule 10: $A + AB = A$ This rule can be proved by applying the distributive law, rule 2, and rule 4 as follows:

$$A + AB = A \cdot 1 + AB = A(1 + B) \quad \text{Factoring (distributive law)}$$

$$= A \cdot 1$$

$$= A$$

$$\text{Rule 2: } (1 + B) = 1$$

$$\text{Rule 4: } A \cdot 1 = A$$

Outcome 1: To understand basics of Digital logic devices

Rule 12: $(A + B)(A + C) = A + BC$ This rule can be proved as follows:

$$\begin{aligned}(A + B)(A + C) &= AA + AC + AB + BC && \text{Distributive law} \\&= A + AC + AB + BC && \text{Rule 7: } AA = A \\&= A(1 + C) + AB + BC && \text{Factoring (distributive law)} \\&= A \cdot 1 + AB + BC && \text{Rule 2: } 1 + C = 1 \\&= A(1 + B) + BC && \text{Factoring (distributive law)} \\&= A \cdot 1 + BC && \text{Rule 2: } 1 + B = 1 \\&= A + BC && \text{Rule 4: } A \cdot 1 = A\end{aligned}$$

Basic rules of Boolean algebra.

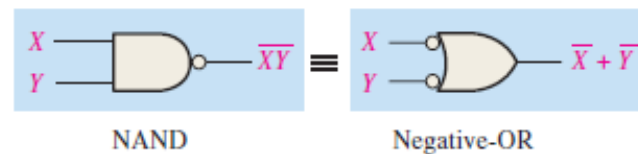
- | | |
|----------------------|-------------------------------|
| 1. $A + 0 = A$ | 7. $A \cdot A = A$ |
| 2. $A + 1 = 1$ | 8. $A \cdot \bar{A} = 0$ |
| 3. $A \cdot 0 = 0$ | 9. $\bar{\bar{A}} = A$ |
| 4. $A \cdot 1 = A$ | 10. $A + AB = A$ |
| 5. $A + A = A$ | 11. $A + \bar{A}B = A + B$ |
| 6. $A + \bar{A} = 1$ | 12. $(A + B)(A + C) = A + BC$ |

Outcome 1: To understand basics of Digital logic devices

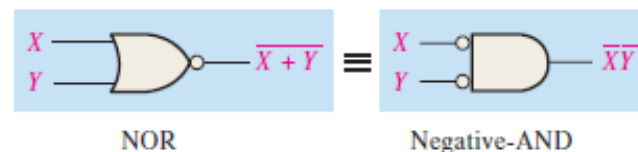
- DeMorgan's Theorems
- First theorem states the complement of a product of variables is equal to the sum of the complements of the variables.
- First theorem in other words: the complement of two or more ANDed variables is equivalent to the OR of the complements of the individual variables. $X'Y' = X'+Y'$
- Second theorem states the complement of a sum of variables is equal to the product of the complements of the variables.
- Second theorem in other words: the complement of two or more ORed variables is equivalent to the AND of the complements of the individual variables. $X'+Y' = X'Y'$

Outcome 1: To understand basics of Digital logic devices

Gate equivalencies and the corresponding truth tables that illustrate DeMorgan's theorems. Notice the equality of the two output columns in each table. This shows that the equivalent gates perform the same logic function.



Inputs		Output	
X	Y	$\overline{XY}$	$\overline{X} + \overline{Y}$
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0



Inputs		Output	
X	Y	$\overline{X} + \overline{Y}$	$\overline{X} \overline{Y}$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

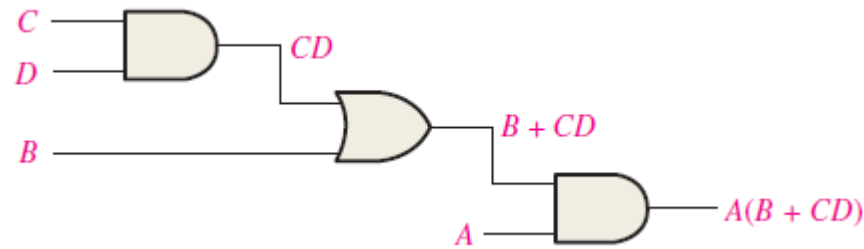
Outcome 1: To understand basics of Digital logic devices

- Each variable in DeMorgan's theorems can also represent a combination of other variables.
- For example if $X = AB+C$ and $Y = A +BC$, then if DeMorgan theorem is applied to $X'Y' = X'+Y'$,
- The resultant is $(AB+C)' + (A+BC)'$
- Further applying DeMorgan's theorem $(AB)'C'+A' (BC)'$
- Which can be further reduced as $(A'+B')C' + A'(B'+C')$
- This result can be simplified further by the use of Boolean rules and laws

$$((A+BC')'+D(E+F'))'$$

Outcome 1: To understand basics of Digital logic devices

- Boolean Expression for a Logic Circuit
- To derive the Boolean expression for a given combinational logic circuit, begin at the left-most inputs and work toward the final output, writing the expression for each gate.



- Constructing a Truth Table for a Logic Circuit
- A truth table that shows the output for all possible values of the input variables can be developed.

Outcome 1: To understand basics of Digital logic devices

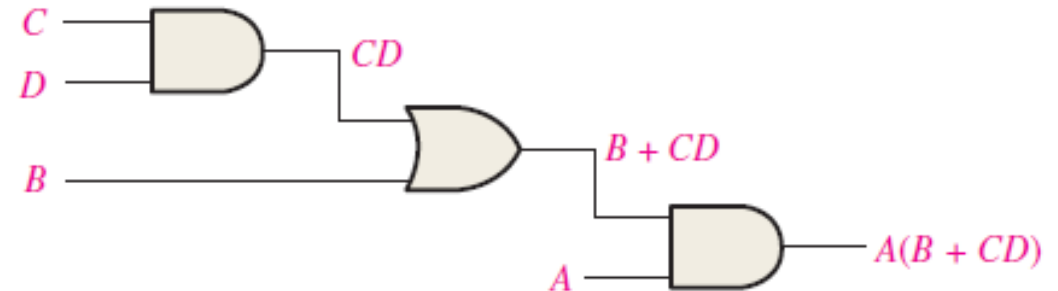
- There are four input variables (A , B , C , and D) and therefore sixteen ($2^4 = 16$) combinations of values are possible.
- First find the values of the variables that make the expression equal to 1, using the rules for Boolean addition and multiplication.
- The expression $A(B+CD)$ is 1 only if $A = 1$ or $(B+CD) = 1$
- In case of $(B+CD)$, If $B = 1$ or $CD = 1$;
- In case of CD , if $C = 1$ and $D = 1$
- So, If $A = 1$ and $B = 1$, then expression is 1 irrespective of C and D
- Also If $A=1$ and $C=1$ and $D=1$, then expression is 1 irrespective of B

Outcome 2

TABLE 4-5

Truth table for the logic circuit in Figure 4-18.

Inputs				Output
<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	$A(B + CD)$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1



Outcome 1: To understand basics of Digital logic devices

- Logic Simplification Using Boolean Algebra
- A logic expression can be reduced to its simplest form or changed to a more convenient form to implement the expression most efficiently using Boolean algebra.

Step 1: Apply the distributive law to the second and third terms in the expression, as follows:

$$AB + AB + AC + BB + BC$$

Step 2: Apply rule 7 ($BB = B$) to the fourth term.

$$AB + AB + AC + B + BC$$

Step 3: Apply rule 5 ($AB + AB = AB$) to the first two terms.

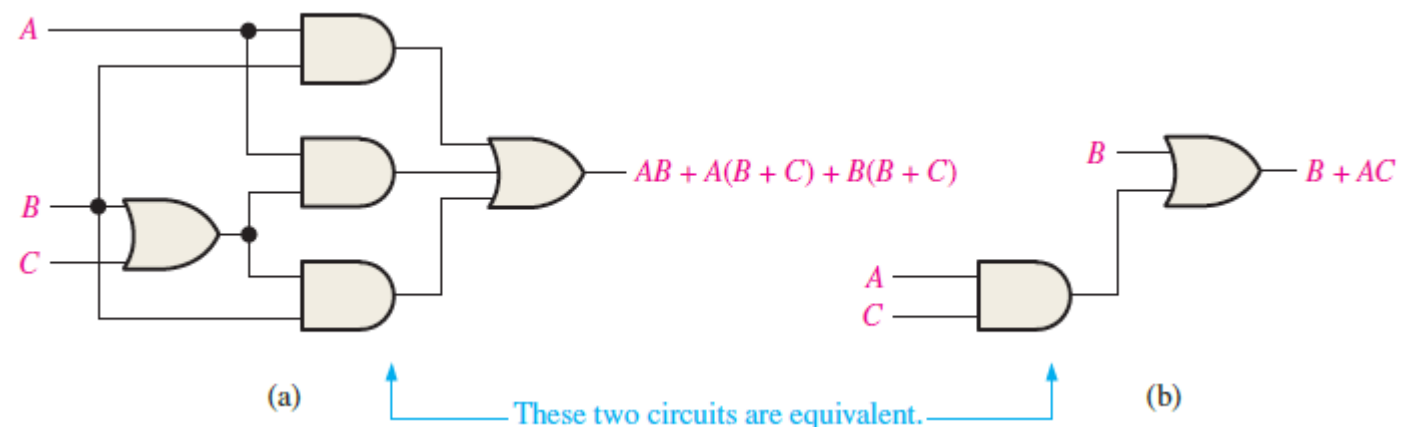
$$AB + AC + B + BC$$

Step 4: Apply rule 10 ($B + BC = B$) to the last two terms.

$$AB + AC + B$$

Step 5: Apply rule 10 ($AB + B = B$) to the first and third terms.

$$B + AC$$



Outcome 1: To understand basics of Digital logic devices

- Standard forms of Boolean expression
- All Boolean expressions, regardless of their form, can be converted into either of two standard forms: the sum-of-products form or the product-of-sums form.
- Standardization makes the evaluation, simplification, and implementation of Boolean expressions much more systematic and easier.

$$AB + ABC$$

$$ABC + CDE + \overline{BCD}$$

$$\overline{AB} + \overline{ABC} + AC$$

- Sum of Product (SOP) Form
- When two or more product terms are summed by Boolean addition

Outcome 1: To understand basics of Digital logic devices

- Domain of a Boolean Expression
- The **domain** of a general Boolean expression is the set of variables contained in the expression in either complemented or uncomplemented form.
- An SOP expression can be implemented with one OR gate and two or more AND gates.
- Any logic expression can be changed into SOP form by applying Boolean algebra techniques.
- $A(B + CD) = AB + ACD$, $((A + B)' + C)' = (A + B)''C' = (A + B)C = AC + BC$

Outcome 1: To understand basics of Digital logic devices

- Standard SOP form
- A *standard SOP expression* is one in which *all* the variables in the domain appear in each product term in the expression.
- Examples seen so far had SOP expressions in which some of the product terms do not contain all of the variables in the domain of the expression.
- A nonstandard SOP expression is converted into standard form using Boolean algebra rule 6 ($A + A' = 1$)
- $AB'C + A'B' + ABC'D$

Outcome 1: To understand basics of Digital logic devices

- **Binary Representation of a Standard Product Term**
- A standard product term is equal to 1 for only one combination of variable values.
- For example, the product term $AB'CD'$ is equal to 1 when $A = 1$, $B = 0$, $C = 1$, $D = 0$, and is 0 for all other combinations of values for the variables.
- **An SOP expression is equal to 1 only if one or more of the product terms in the expression is equal to 1.**

Outcome 1: To understand basics of Digital logic devices

- Product of Sum (POS) Form
- When two or more sum terms are multiplied

$$\begin{aligned} &(\bar{A} + B)(A + \bar{B} + C) \\ &(\bar{A} + \bar{B} + \bar{C})(C + \bar{D} + E)(\bar{B} + C + D) \\ &(A + B)(A + \bar{B} + C)(\bar{A} + C) \end{aligned}$$

- Implementing a POS expression simply requires ANDing the outputs of two or more OR gates.
- Standard POS form
- A *standard POS expression* is one in which *all* the variables in the domain appear in each sum term in the expression.

Outcome 1: To understand basics of Digital logic devices

- A nonstandard POS expression is converted into standard form using Boolean algebra rule 8 ($AA' = 0$)
- $(A + B' + C)(B' + C + D')(A + B' + C' + D)$
- **Binary Representation of a Standard Sum Term**
- A standard sum term is equal to 0 for only one combination of variable values. For example, the sum term $A + B' + C + D'$ is 0 when $A = 0$, $B = 1$, $C = 0$, and $D = 1$, and is 1 for all other combinations of values for the variables.
- **A POS expression is equal to 0 only if one or more of the sum terms in the expression is equal to 0.**

Outcome 1: To understand basics of Digital logic devices

- Converting Standard SOP to Standard POS
- **Step 1:** Evaluate each product term in the SOP expression. That is, determine the binary numbers that represent the product terms.
- **Step 2:** Determine all of the binary numbers not included in the evaluation in Step 1.
- **Step 3:** Write the equivalent sum term for each binary number from Step 2 and express in POS form.

Convert the following SOP expression to an equivalent POS expression:

$$\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}C + ABC$$

Solution

The evaluation is as follows:

$$000 + 010 + 011 + 101 + 111$$

Since there are three variables in the domain of this expression, there are a total of eight (2^3) possible combinations. The SOP expression contains five of these combinations, so the POS must contain the other three which are 001, 100, and 110. Remember, these are the binary values that make the sum term 0. The equivalent POS expression is

$$(A + B + \bar{C})(\bar{A} + B + C)(\bar{A} + \bar{B} + C)$$

Outcome 1: To understand basics of Digital logic devices

- Karnaugh map (K-map)
- provides a systematic method for simplifying Boolean expressions and, if properly used, will produce the simplest SOP or POS expression possible, known as the minimum expression.
- A **Karnaugh map** is similar to a truth table because it presents all of the possible values of input variables and the resulting output for each value.
- Instead of being organized into columns and rows like a truth table, the Karnaugh map is an array of **cells** in which each cell represents a binary value of the input variables.

Outcome 1: To understand basics of Digital logic devices

- The number of cells in a Karnaugh map, as well as the number of rows in a truth table, is equal to the total number of possible input variable combinations. For three variables, the number of cells is $2^3 = 8$. For four variables, the number of cells is $2^4 = 16$.
- 3 Variable K-map and 4 Variable K-map

AB \ C	C	
	0	1
00		
01		
11		
10		

(a)

AB \ C	C	
	0	1
00	$\bar{A}\bar{B}\bar{C}$	$\bar{A}\bar{B}C$
01	$\bar{A}B\bar{C}$	$\bar{A}BC$
11	$AB\bar{C}$	ABC
10	$A\bar{B}\bar{C}$	$A\bar{B}C$

(b)

AB \ CD	CD			
	00	01	11	10
00				
01				
11				
10				

(a)

AB \ CD	CD			
	00	01	11	10
00	$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}C\bar{D}$	$\bar{A}\bar{B}CD$
01	$\bar{A}B\bar{C}\bar{D}$	$\bar{A}B\bar{C}D$	$\bar{A}BC\bar{D}$	$\bar{A}BCD$
11	$AB\bar{C}\bar{D}$	$AB\bar{C}D$	$ABC\bar{D}$	$ABCD$
10	$A\bar{B}\bar{C}\bar{D}$	$A\bar{B}\bar{C}D$	$A\bar{B}C\bar{D}$	$A\bar{B}CD$

(b)

Outcome 1: To understand basics of Digital logic devices

- The cells in a Karnaugh map are arranged so that there is only a single-variable change between adjacent cells. **Adjacency** is defined by a single-variable change.
- In the 3-variable map the 010 cell is adjacent to the 000 cell, the 011 cell, and the 110 cell. The 010 cell is not adjacent to the 001 cell, the 111 cell, the 100 cell, or the 101 cell.
- A cell is not adjacent to the cells that diagonally touch any of its corners.
- Also, the cells in the top row are adjacent to the corresponding cells in the bottom row and the cells in the outer left column are adjacent to the corresponding cells in the outer right column. This is called “wrap-around” adjacency

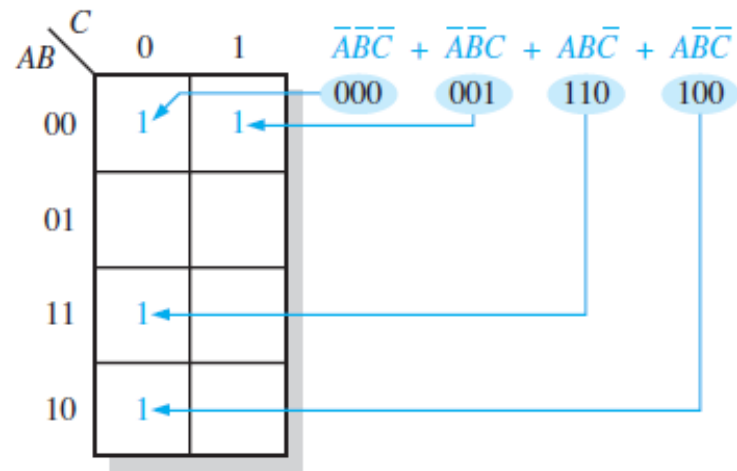
Outcome 1: To understand basics of Digital logic devices

- Mapping a Standard SOP Expression

Step 1: Determine the binary value of each product term in the standard SOP expression. After some practice, you can usually do the evaluation of terms mentally.

Step 2: As each product term is evaluated, place a 1 on the Karnaugh map in the cell having the same value as the product term.

$$A'B'C + A'BC' + ABC' + ABC$$



Outcome 1: To understand basics of Digital logic devices

- Mapping a Nonstandard SOP Expression

Map the following SOP expression on a Karnaugh map: $\bar{A} + A\bar{B} + ABC$.

Solution

The SOP expression is obviously not in standard form because each product term does not have three variables. The first term is missing two variables, the second term is missing one variable, and the third term is standard. First expand the terms numerically as follows:

$$\begin{array}{l} \bar{A} + A\bar{B} + ABC \\ 000 \quad 100 \quad 110 \\ 001 \quad 101 \\ 010 \\ 011 \end{array}$$

Map each of the resulting binary values by placing a 1 in the appropriate cell of the 3-variable Karnaugh map in Figure 4–31.

		C	
		0	1
AB	00	1	1
	01	1	1
	11	1	
	10	1	1

Outcome 1: To understand basics of Digital logic devices

- Karnaugh Map Simplification of SOP Expressions
- **Grouping the 1s**
- A group must contain either 1, 2, 4, 8, or 16 cells, which are all powers of two. In the case of a 3-variable map, $2^3 = 8$ cells is the maximum group.
- Each cell in a group must be adjacent to one or more cells in that same group, but all cells in the group do not have to be adjacent to each other.
- Always include the largest possible number of 1s in a group in accordance with rule 1.
- Each 1 on the map must be included in at least one group. The 1s already in a group can be included in another group as long as the overlapping groups include noncommon 1s.

Outcome 1: To understand basics of Digital logic devices

AB \ C	C	
	0	1
00	1	
01		1
11	1	1
10		

AB \ C	C	
	0	1
00	1	1
01	1	
11		1
10	1	1

AB \ CD	CD			
	00	01	11	10
00	1	1		
01	1	1	1	1
11				
10		1	1	

AB \ CD	CD			
	00	01	11	10
00	1			1
01	1	1		1
11	1	1		1
10	1		1	1

(a)

AB \ C	C	
	0	1
00	1	
01		1
11	1	1
10		

(b)

AB \ C	C	
	0	1
00	1	1
01	1	
11		1
10	1	1

Wrap-around adjacency

(c)

AB \ CD	CD			
	00	01	11	10
00	1	1		
01	1	1	1	1
11				
10		1	1	

(d)

AB \ CD	CD			
	00	01	11	10
00	1			1
01	1	1		1
11	1	1		1
10	1		1	1

Wrap-around adjacency

Outcome 1: To understand basics of Digital logic devices

- **Determining the Minimum SOP Expression from the Map**
- Group the cells that have 1s. Each group of cells containing 1s creates one product term composed of all variables that occur in only one form (either uncomplemented or complemented) within the group. Variables that occur both uncomplemented and complemented within the group are eliminated. These are called *contradictory variables*.
- Determine the minimum product term for each group.
- When all the minimum product terms are derived from the Karnaugh map, they are summed to form the minimum SOP expression.

Outcome 1: To understand basics of Digital logic devices

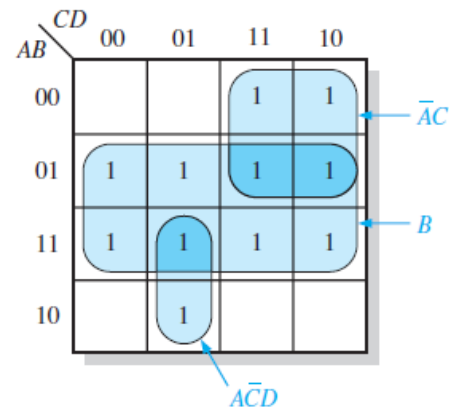


FIGURE 4-35

Solution

Eliminate variables that are in a grouping in both complemented and uncomplemented forms. In Figure 4-35, the product term for the 8-cell group is B because the cells within that group contain both A and $\bar{A}$, C and $\bar{C}$, and D and $\bar{D}$, which are eliminated. The 4-cell group contains B , $\bar{B}$, D , and $\bar{D}$, leaving the variables $\bar{A}$ and C , which form the product term $\bar{A}C$. The 2-cell group contains B and $\bar{B}$, leaving variables A , $\bar{C}$, and D which form the product term $A\bar{C}D$. Notice how overlapping is used to maximize the size of the groups. The resulting minimum SOP expression is the sum of these product terms:

$$B + \bar{A}C + A\bar{C}D$$

$$AB'C + A'BC + A'B'C + A'B'C' + AB'C'$$

$$\text{Solution } B' + A'C$$

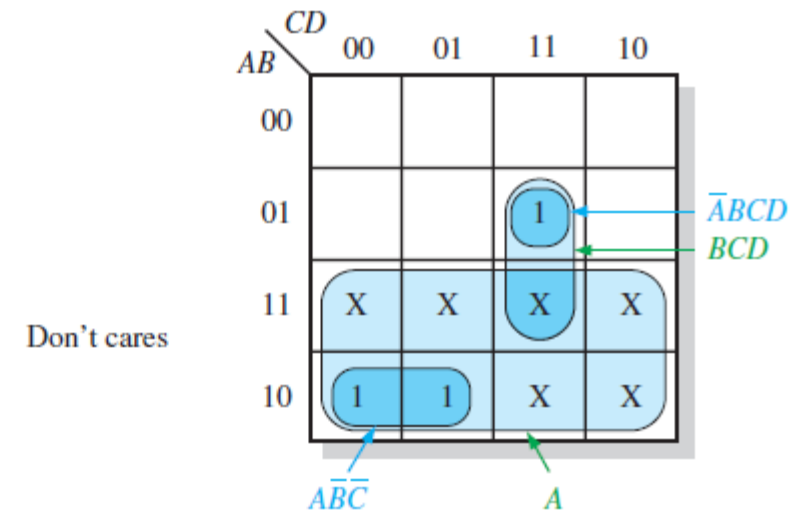
Outcome 1: To understand basics of Digital logic devices

- “Don’t Care” Conditions
- Sometimes a situation arises in which some input variable combinations are not allowed.
- For example, recall that in the BCD code, there are six invalid combinations: 1010, 1011, 1100, 1101, 1110, and 1111.

Inputs				Output
A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

(a) Truth table

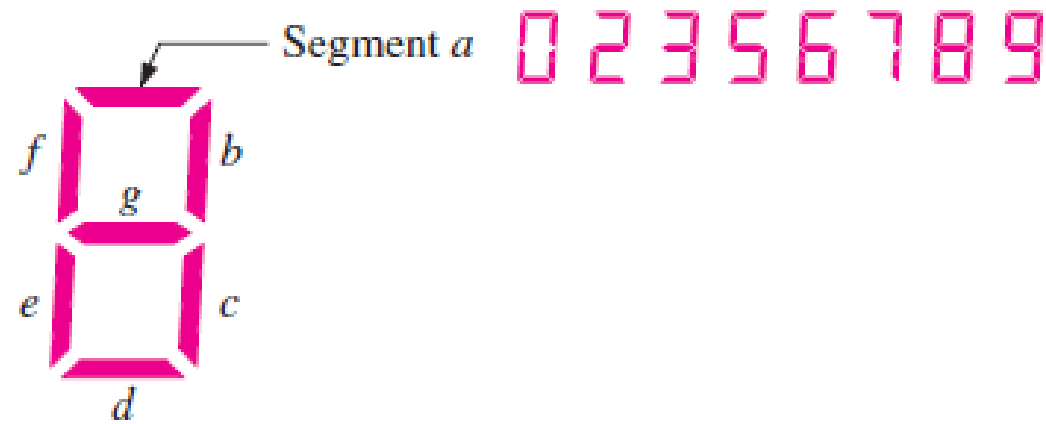
The truth table describes a logic function that has a 1 output only when the BCD code for 7, 8, or 9 is present on the inputs.



(b) Without “don’t cares” $Y = \bar{A}\bar{B}\bar{C} + \bar{A}BCD$
With “don’t cares” $Y = A + BCD$

Outcome 2 & 4

In a 7-segment display, each of the seven segments is activated for various digits. For example, segment *a* is activated for the digits 0, 2, 3, 5, 6, 7, 8, and 9, as illustrated in Figure 4–41. Since each digit can be represented by a BCD code, derive an SOP expression for segment *a* using the variables *ABCD* and then minimize the expression using a Karnaugh map.

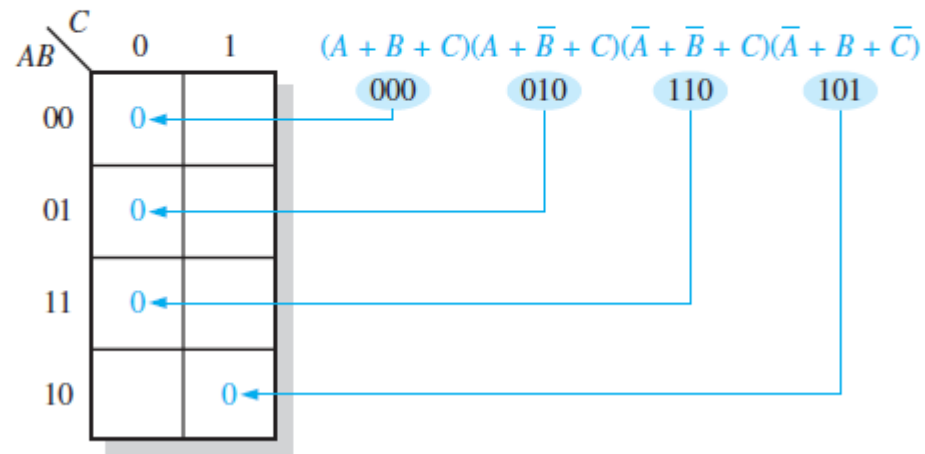


Outcome 1: To understand basics of Digital logic devices

- Karnaugh Map POS Minimization

Step 1: Determine the binary value of each sum term in the standard POS expression. This is the binary value that makes the term equal to 0.

Step 2: As each sum term is evaluated, place a 0 on the Karnaugh map in the corresponding cell.



Outcome 1: To understand basics of Digital logic devices

- Karnaugh Map Simplification of POS Expressions

Use a Karnaugh map to minimize the following standard POS expression:

$$(A + B + C)(A + B + \overline{C})(A + \overline{B} + C)(A + \overline{B} + \overline{C})(\overline{A} + \overline{B} + C)$$

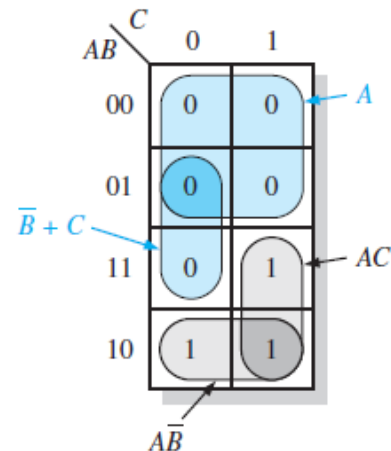
Also, derive the equivalent SOP expression.

Solution

The combinations of binary values of the expression are

$$(0 + 0 + 0)(0 + 0 + 1)(0 + 1 + 0)(0 + 1 + 1)(1 + 1 + 0)$$

Map the standard POS expression and group the cells as shown in Figure 4-45.



Notice how the 0 in the 110 cell is included into a 2-cell group by utilizing the 0 in the 4-cell group. The sum term for each blue group is shown in the figure and the resulting minimum POS expression is

$$A(\overline{B} + C)$$

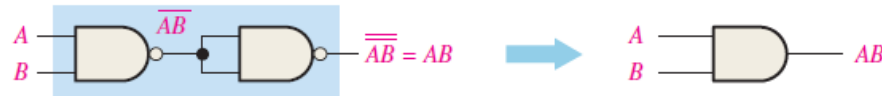
Converting Between POS and SOP Using the Karnaugh Map

Outcome 1: To understand basics of Digital logic devices

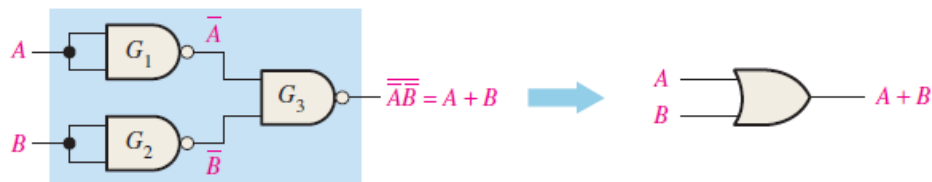
NAND and NOR gate as universal gates



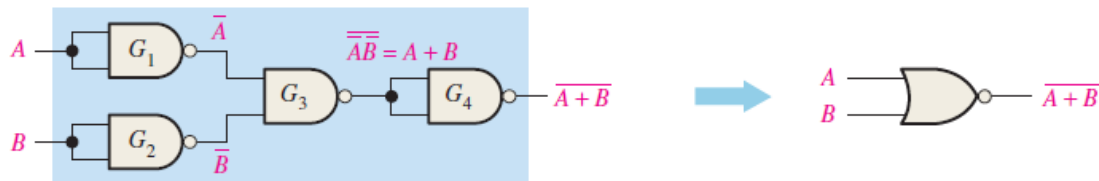
(a) One NAND gate used as an inverter



(b) Two NAND gates used as an AND gate



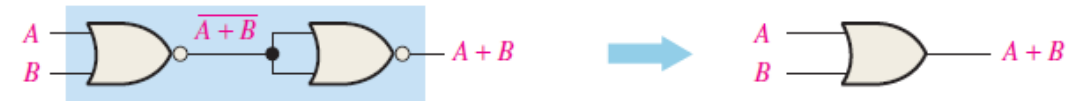
(c) Three NAND gates used as an OR gate



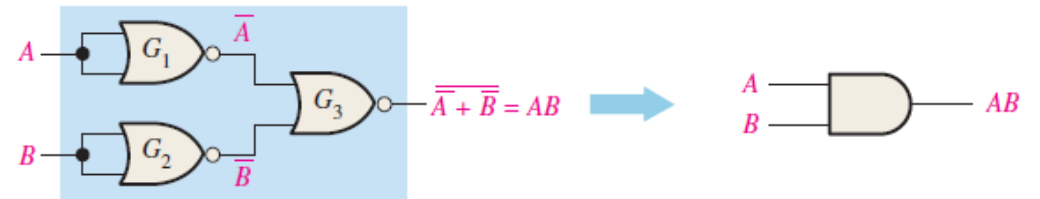
(d) Four NAND gates used as a NOR gate



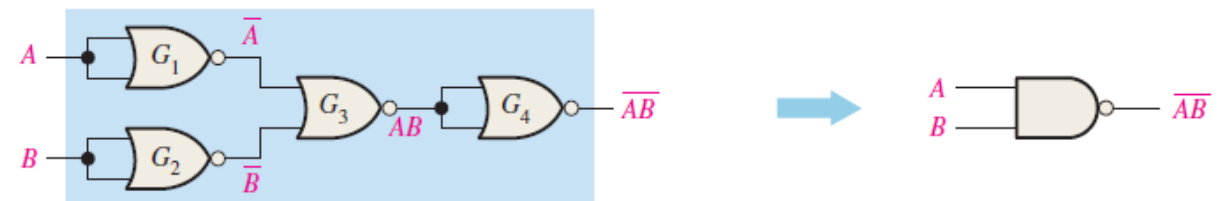
(a) One NOR gate used as an inverter



(b) Two NOR gates used as an OR gate



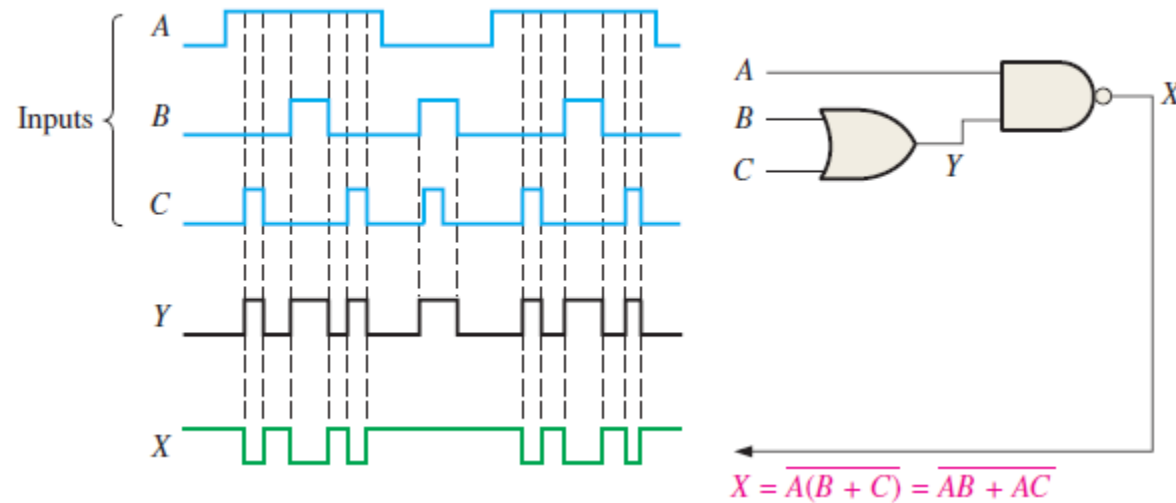
(c) Three NOR gates used as an AND gate



(d) Four NOR gates used as a NAND gate

Outcome 1: To understand basics of Digital logic devices

Determine the final output waveform X for the circuit in Figure 5–31, with input waveforms A , B , and C as shown.



After Mid Term

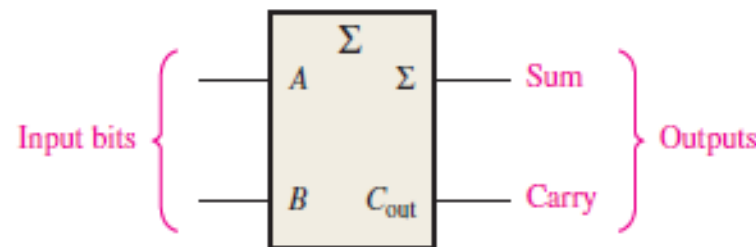
Outcome 1: To understand basics of Digital logic devices

- When logic gates are connected together to produce a specified output for certain specified combinations of input variables, with no storage involved, the resulting circuit is in the category of **combinational logic**.
- SOP and POS implementations, are basic forms of combinational logic.
- In combinational logic, the output level is at all times dependent on the combination of input levels.

Outcome 1: To understand basics of Digital logic devices

- Fixed function combinational logic circuits such as half and full adders, half and full subtractor, decoders, encoders, multiplexers (data selectors), and demultiplexers
- Adders are important in computers and also in other types of digital systems in which numerical data are processed.
- A half-adder adds two bits and produces a sum and an output carry.

0	+	0	=	0
0	+	1	=	1
1	+	0	=	1
1	+	1	=	10



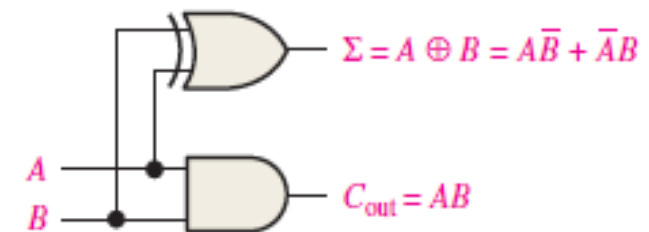
Half-adder truth table.

A	B	C_{out}	Σ
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Σ = sum

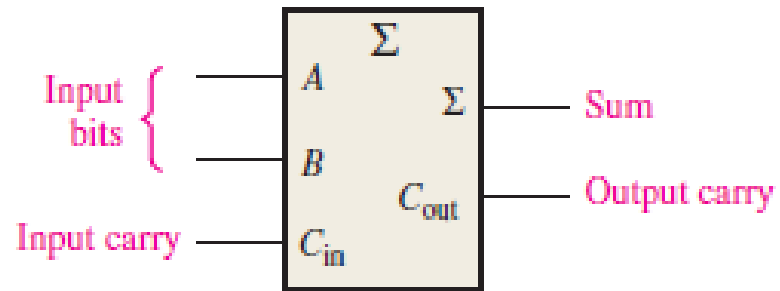
C_{out} = output carry

A and B = input variables (operands)



Outcome 1: To understand basics of Digital logic devices

- The full-adder accepts two input bits and an input carry and generates a sum output and an output carry.
- The basic difference between a full-adder and a half-adder is that the full-adder accepts an input carry.



Full-adder truth table.

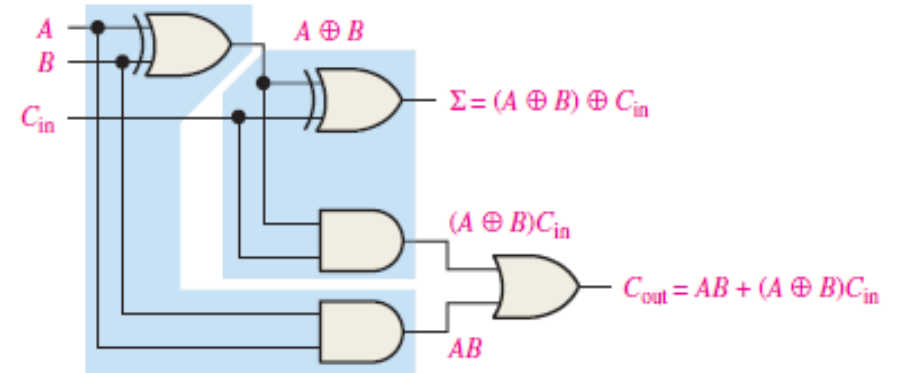
A	B	C_{in}	C_{out}	Σ
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

C_{in} = input carry, sometimes designated as CI

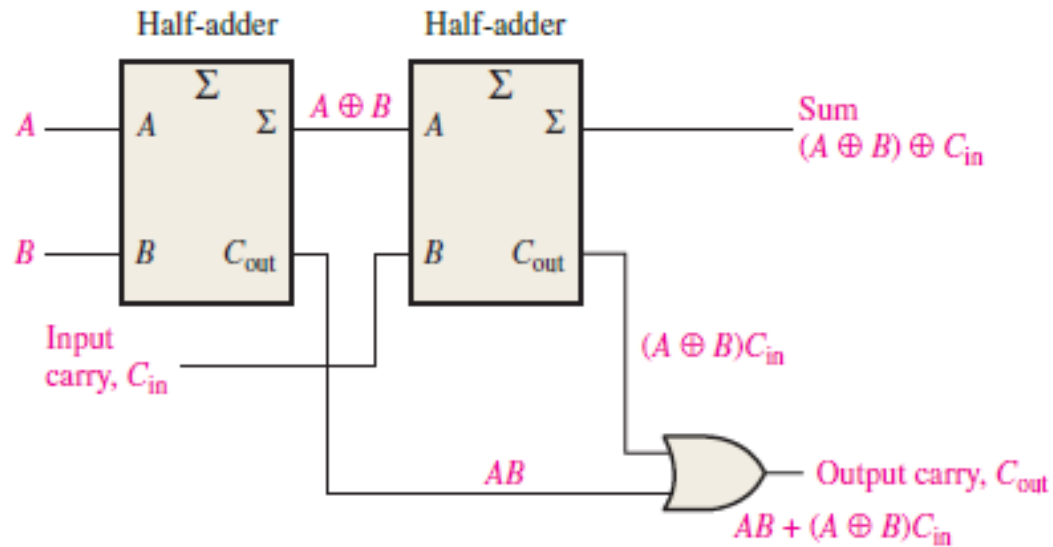
C_{out} = output carry, sometimes designated as CO

Σ = sum

A and B = input variables (operands)

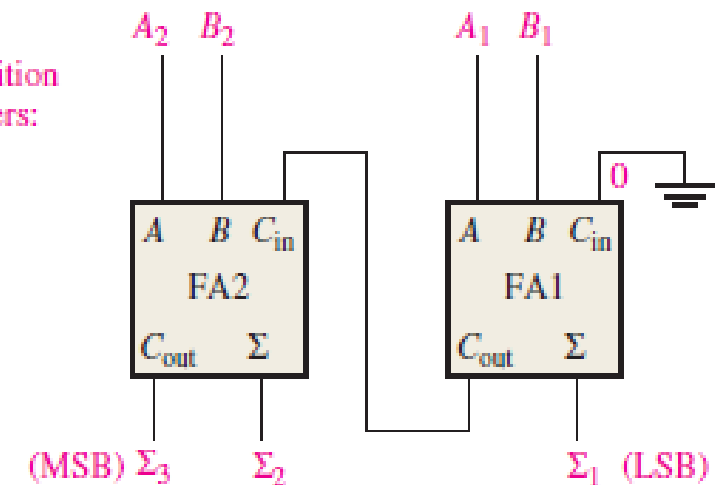


Outcome 1: To understand basics of Digital logic devices



General format, addition of two 2-bit numbers:

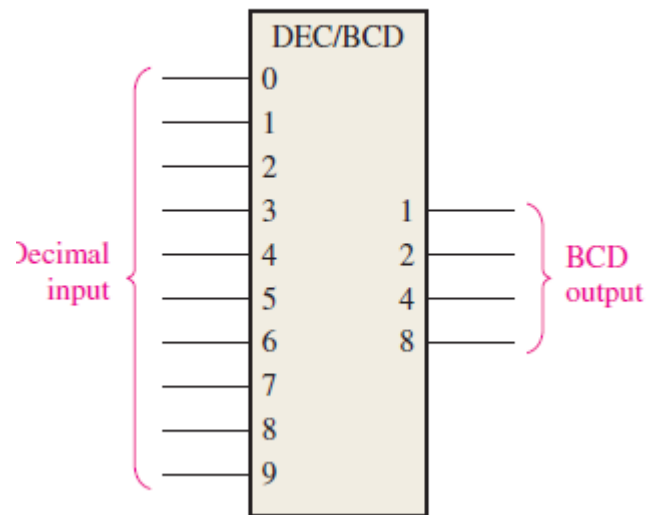
$$\begin{array}{r} A_2A_1 \\ + B_2B_1 \\ \hline \Sigma_3\Sigma_2\Sigma_1 \end{array}$$



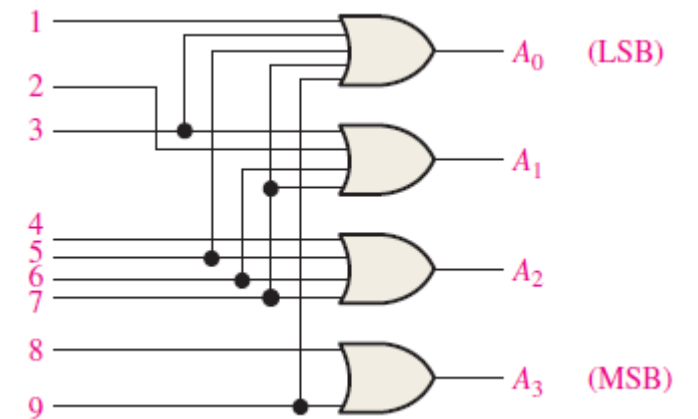
74LS283 is a four bit full adder

Outcome 1: To understand basics of Digital logic devices

- Encoders
- The process of converting from familiar symbols or numbers to a coded format is called *encoding*.



Decimal Digit	BCD Code			
	A_3	A_2	A_1	A_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

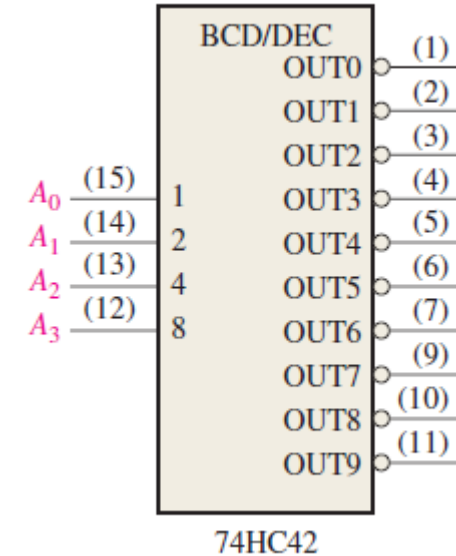


Outcome 1: To understand basics of Digital logic devices

- Decoder
- BCD-to-Decimal Decoder or *4-line-to-10-line decoder*

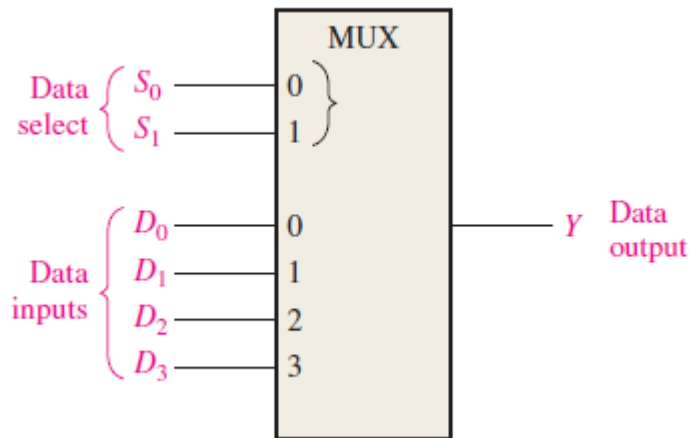
BCD decoding functions.

Decimal Digit	BCD Code				Decoding Function
	A_3	A_2	A_1	A_0	
0	0	0	0	0	$\overline{A_3}\overline{A_2}\overline{A_1}\overline{A_0}$
1	0	0	0	1	$\overline{A_3}\overline{A_2}\overline{A_1}A_0$
2	0	0	1	0	$\overline{A_3}\overline{A_2}A_1\overline{A_0}$
3	0	0	1	1	$\overline{A_3}\overline{A_2}A_1A_0$
4	0	1	0	0	$\overline{A_3}A_2\overline{A_1}\overline{A_0}$
5	0	1	0	1	$\overline{A_3}A_2\overline{A_1}A_0$
6	0	1	1	0	$\overline{A_3}A_2A_1\overline{A_0}$
7	0	1	1	1	$\overline{A_3}A_2A_1A_0$
8	1	0	0	0	$A_3\overline{A_2}\overline{A_1}\overline{A_0}$
9	1	0	0	1	$A_3\overline{A_2}\overline{A_1}A_0$



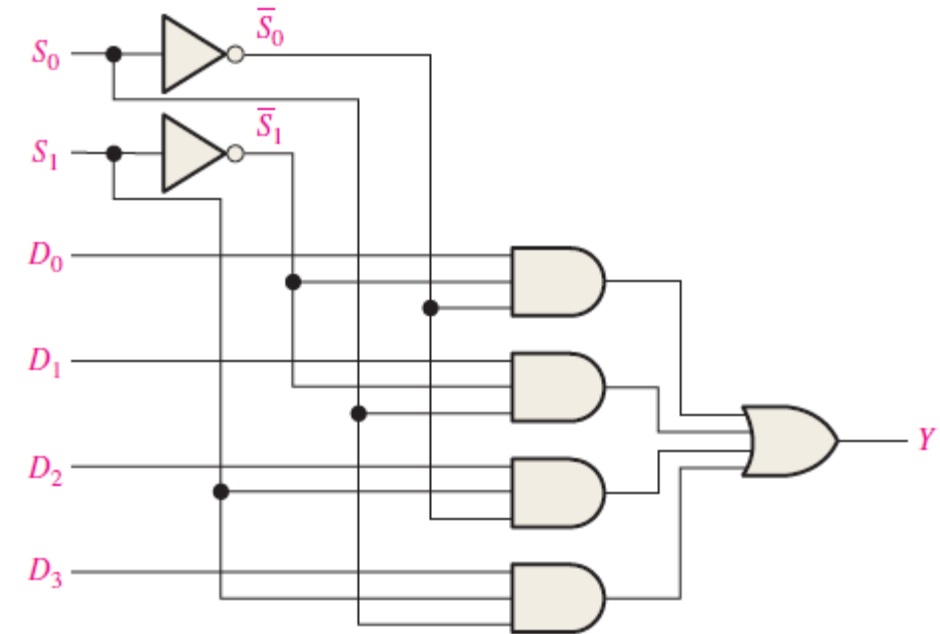
Outcome 1: To understand basics of Digital logic devices

- Multiplexers (Data Selectors)



Data selection for a 1-of-4-multiplexer.

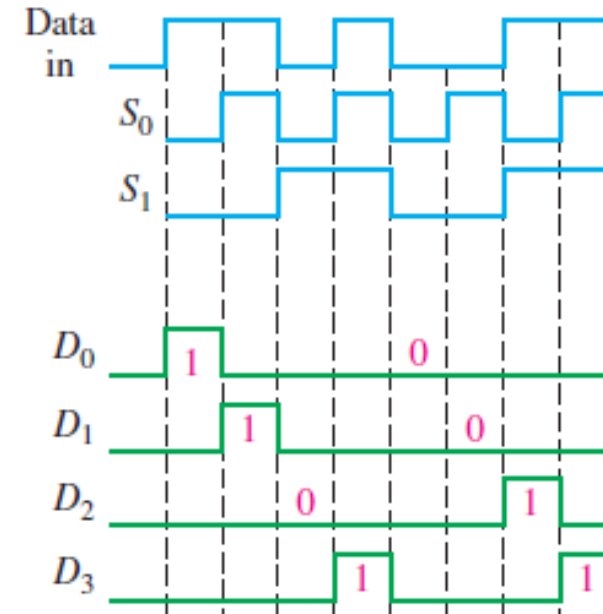
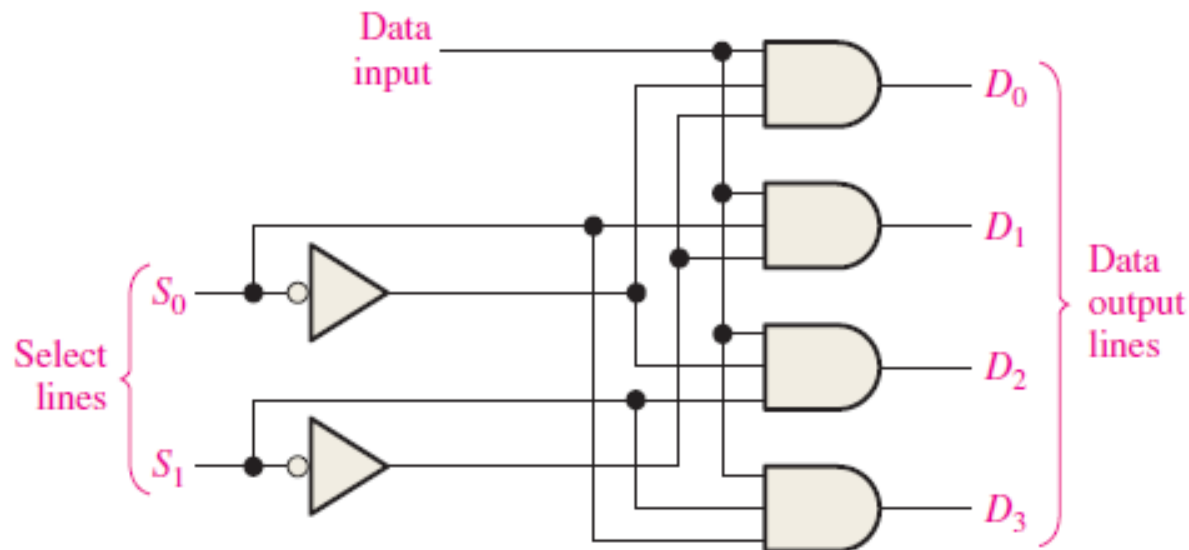
Data-Select Inputs		Input Selected
S_1	S_0	
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3



74HC153 dual 4:1 mux IC

Outcome 1: To understand basics of Digital logic devices

- **demultiplexer (DEMUX)**



Outcome 2 & 4

22. A 7-segment decoder/driver drives the display in Figure 6–78. If the waveforms are applied as indicated, determine the sequence of digits that appears on the display.

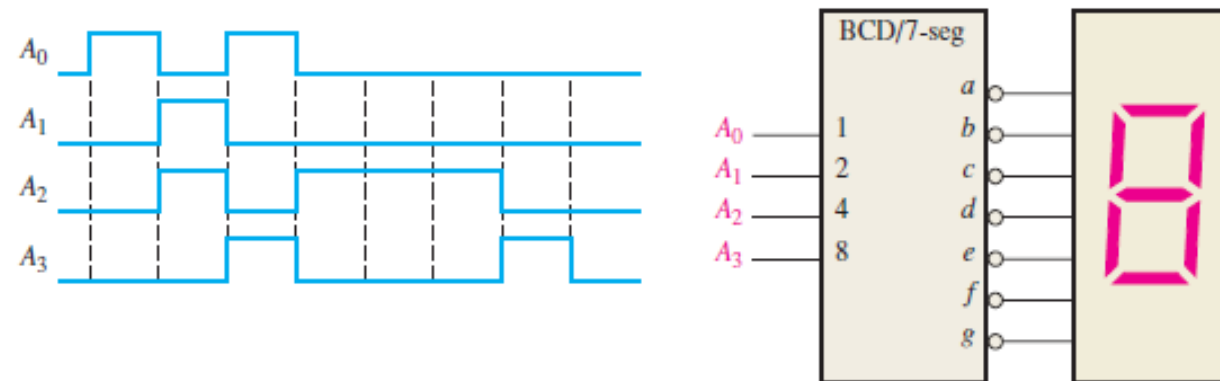
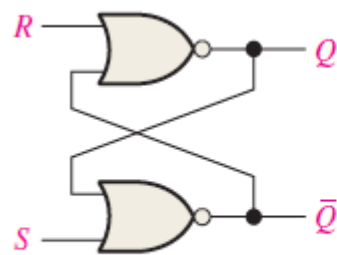


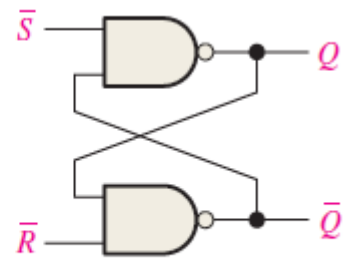
FIGURE 6–78

Outcome 1: To understand basics of Digital logic devices

- Latches
- a type of temporary storage device that has two stable states (bistable)
- S-R (SET-RESET) Latch
- An active-HIGH input S-R (SET-RESET) latch is formed with two cross-coupled NOR gates
- an active-LOW input S@R latch is formed with two cross-coupled NAND gates



(a) Active-HIGH input S-R latch



(b) Active-LOW input $\bar{S}$ - $\bar{R}$ latch

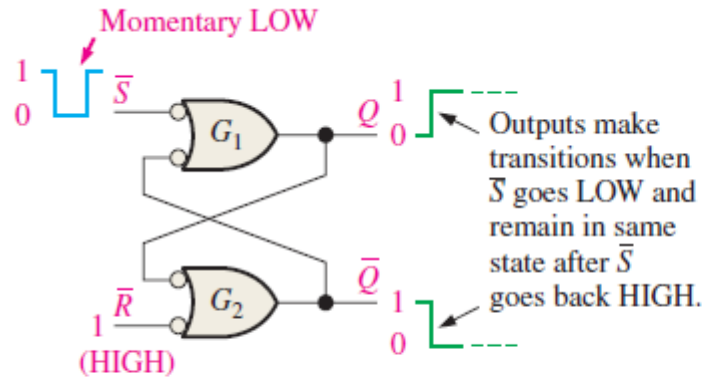
Outcome 1: To understand basics of Digital logic devices

- Notice that the output of each gate is connected to an input of the opposite gate. This produces the regenerative **feedback** that is characteristic of all latches and flipflops
- Assume that both inputs (S and R) and the Q output are HIGH, which is the normal latched state.
- When the Q output is HIGH, the latch is in the **SET** state.
 - It will remain in this state indefinitely until a LOW is temporarily applied to the R input.
- When the Q output is LOW, the latch is in the **RESET** state.
 - The latch remains indefinitely in the RESET state until a momentary LOW is applied to the S input.

Outcome 1: To understand basics of Digital logic devices

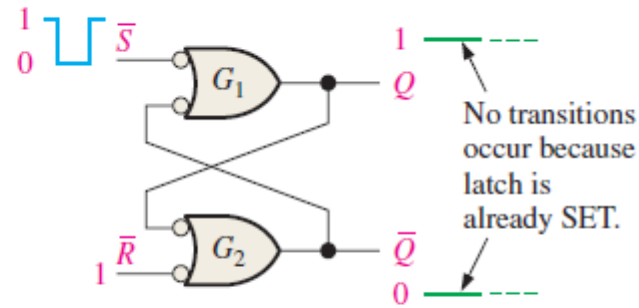
- In normal operation, the outputs of a latch are always complements of each other.
- **When Q is HIGH, $\bar{Q}$ is LOW, and when Q is LOW, $\bar{Q}$ is HIGH.**
- An invalid condition in the operation of an active-LOW input S-R latch occurs when LOWs are applied to both S and R at the same time. As long as the LOW levels are simultaneously held on the inputs, both the Q and $\bar{Q}$ outputs are forced HIGH, thus violating the basic complementary operation of the outputs.
- Also, if the LOWs are released simultaneously, both outputs will attempt to go LOW. Since there is always some small difference in the propagation delay time of the gates, one of the gates will dominate in its transition to the LOW output state. This, in turn, forces the output of the slower gate to remain HIGH. In this situation, you cannot reliably predict the next state of the latch.

Outcome 1: To understand basics of Digital logic devices

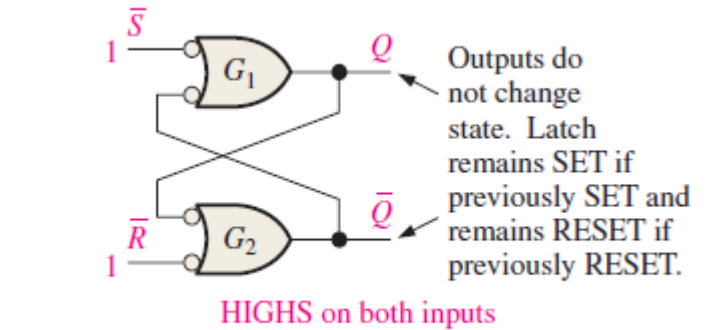


Latch starts out RESET ($Q = 0$).

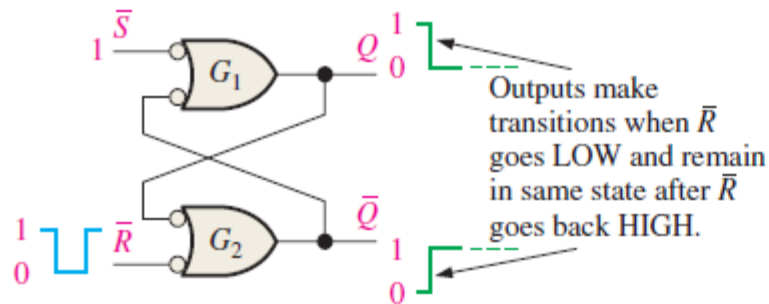
(a) Two possibilities for the SET operation



Latch starts out SET ($Q = 1$).

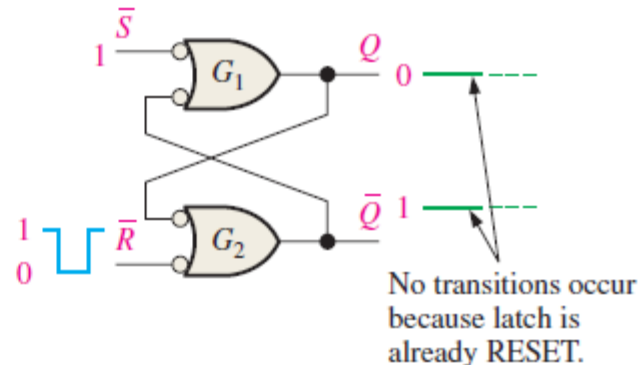


(c) No-change condition

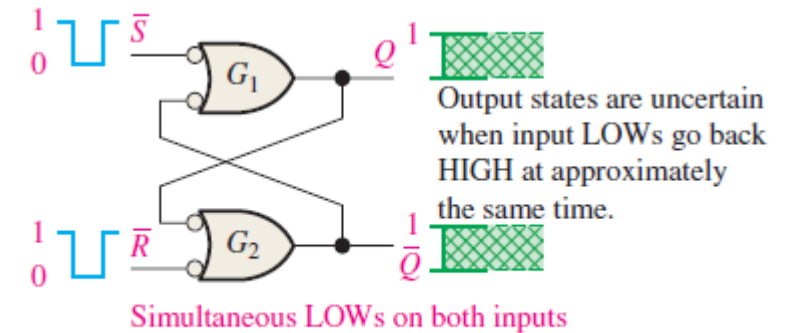


Latch starts out SET ($Q = 1$).

(b) Two possibilities for the RESET operation



Latch starts out RESET ($Q = 0$).

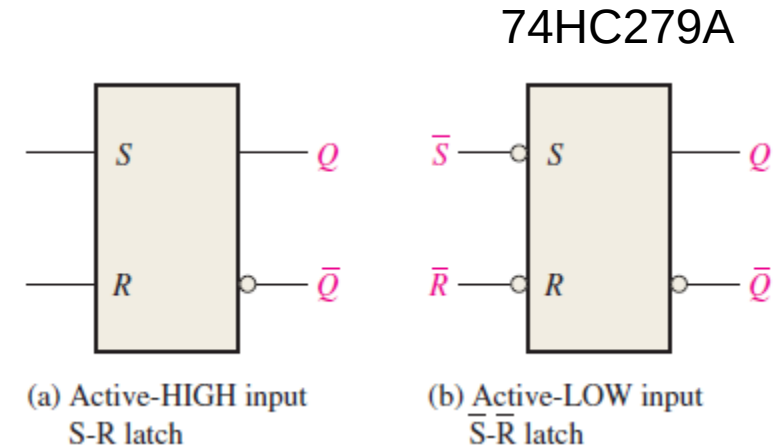


(d) Invalid condition

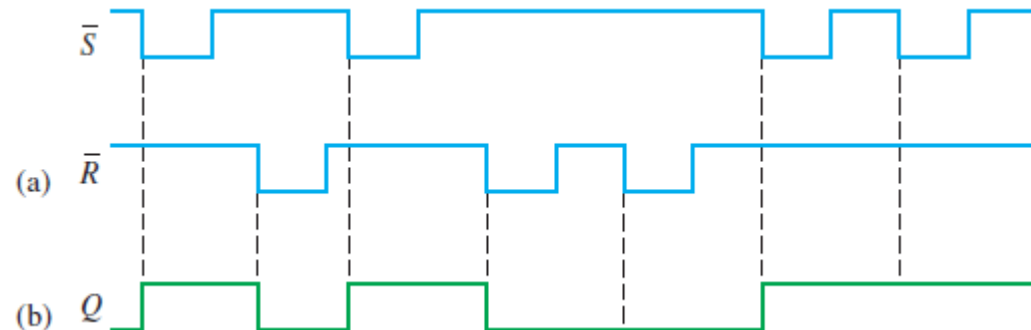
Outcome 1: To understand basics of Digital logic devices

Truth table for an active-LOW input $\bar{S}$ - $\bar{R}$ latch.

Inputs		Outputs		Comments
$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$	
1	1	NC	NC	No change. Latch remains in present state.
0	1	1	0	Latch SET.
1	0	0	1	Latch RESET.
0	0	1	1	Invalid condition



Logic Symbol

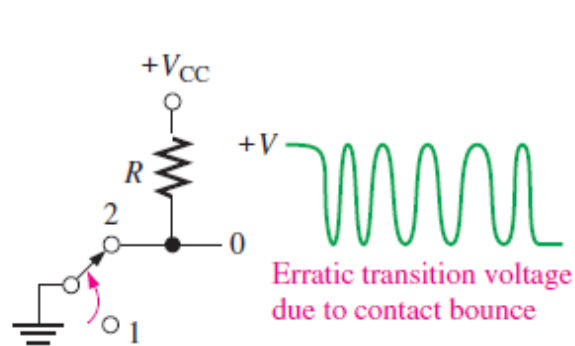


If the S and R waveforms in Figure are applied to the inputs of the S-R latch, determine the waveform that will be observed on the Q output. Assume that Q is initially LOW.

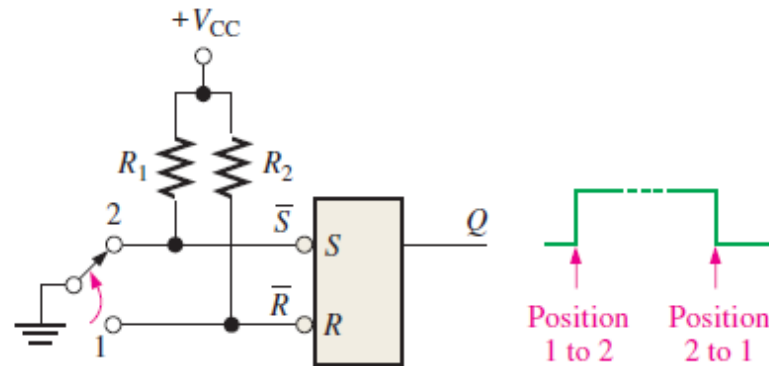
Outcome 2 & 4

74HC279A is a quad S@R latch

- **The Latch as a Contact-Bounce Eliminator**
- an application of an S-R latch is in the elimination of mechanical switch contact “bounce.” When the pole of a switch strikes the contact upon switch closure, it physically vibrates or bounces several times before finally making a solid contact. Although these bounces are very short in duration, they produce voltage spikes that are often not acceptable in a digital system.



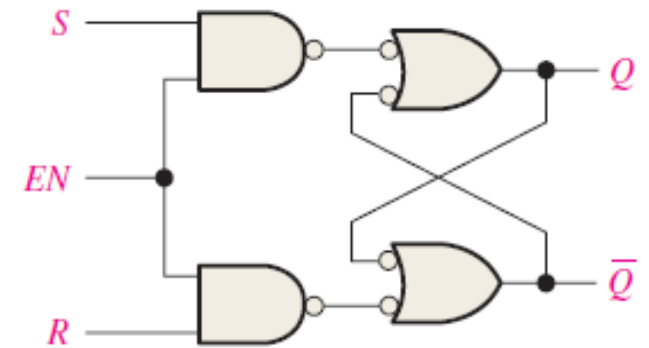
(a) Switch contact bounce



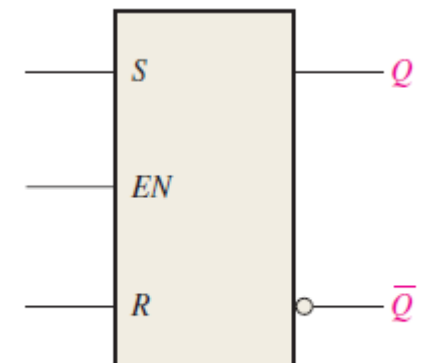
(b) Contact-bounce eliminator circuit

Outcome 1: To understand basics of Digital logic devices

- Gated S-R Latch
- The S and R inputs control the state to which the latch will go when a HIGH level is applied to the EN input.
- The latch will not change until EN is HIGH; but as long as it remains HIGH, the output is controlled by the state of the S and R inputs.
- The gated latch is a *level-sensitive* device.
- In this circuit, the invalid state occurs when both S and R are simultaneously HIGH and EN is also HIGH.



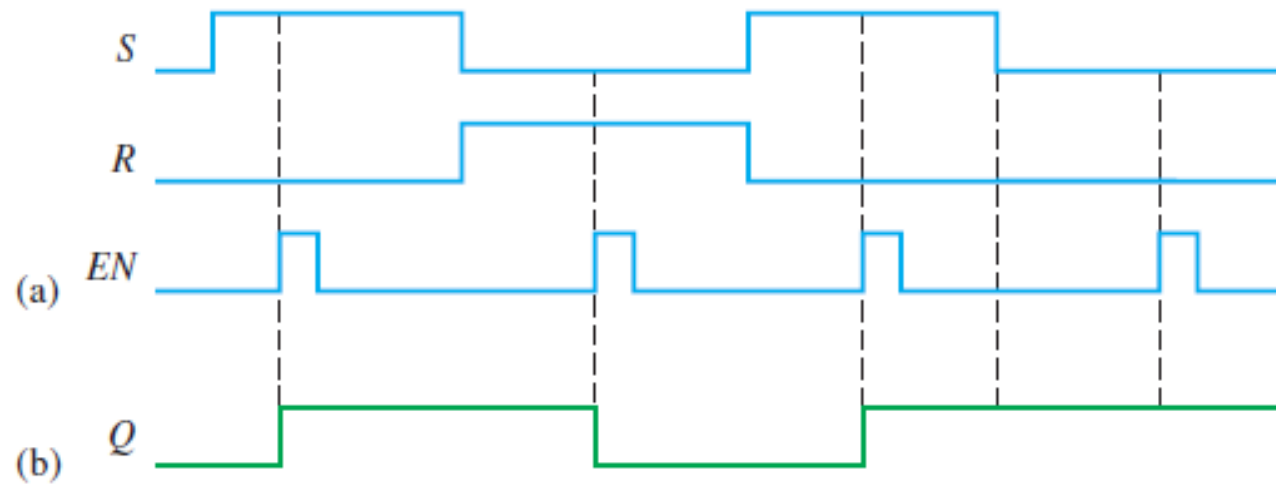
(a) Logic diagram



(b) Logic symbol

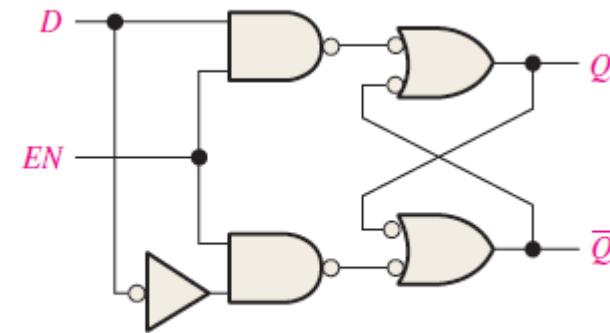
Outcome 2

Determine the Q output waveform if the inputs shown in Figure are applied to a gated S-R latch that is initially RESET.

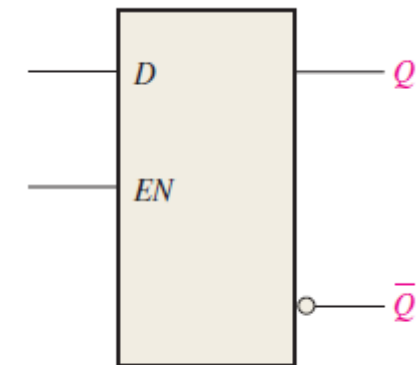


Outcome 1: To understand basics of Digital logic devices

- Gated D Latch
- It differs from the S-R latch because it has only one input in addition to EN . This input is called the D (data) input.
- When the D input is HIGH and the EN input is HIGH, the latch will set. When the D input is LOW and EN is HIGH, the latch will reset.
- Stated another way, the output Q follows the input D when EN is HIGH.



(a) Logic diagram



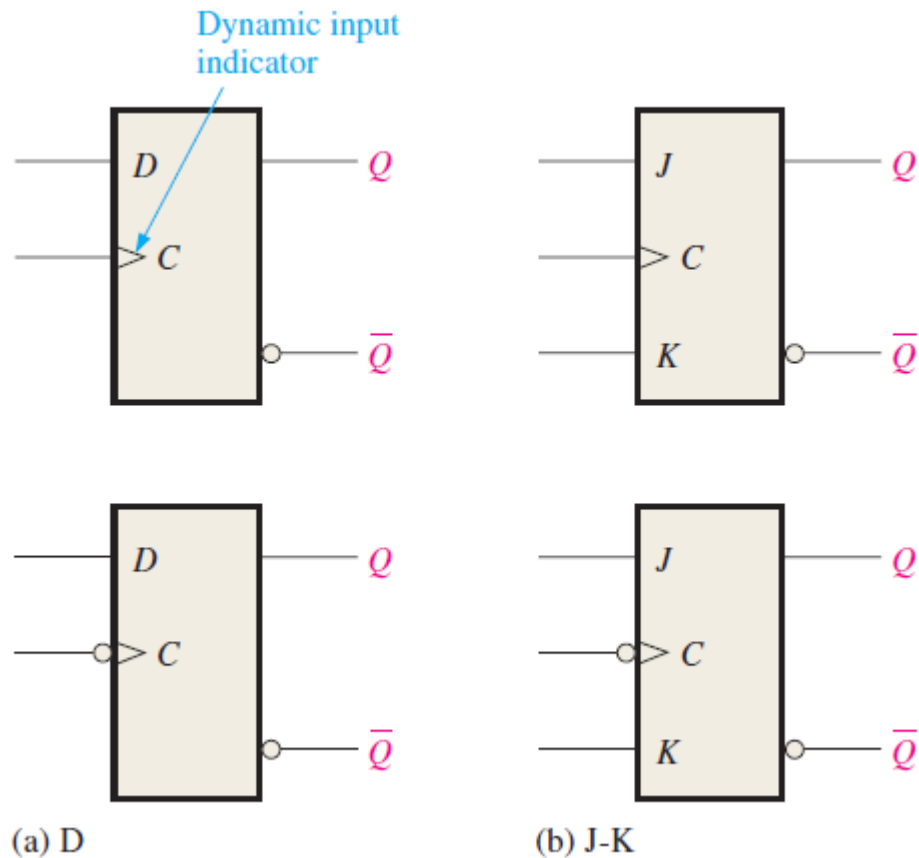
(b) Logic symbol

74HC75

Outcome 1: To understand basics of Digital logic devices

- Flip Flop
- Flip-flops are synchronous bistable devices, also known as bistable multivibrators.
- In this case, the term synchronous means that the output changes state only at a specified point (leading or trailing edge) on the triggering input called the clock (CLK), which is designated as a control input, C; that is, changes in the output occur in synchronization with the clock.
- Flip-flops are edge-triggered or edge-sensitive whereas gated latches are level-sensitive.

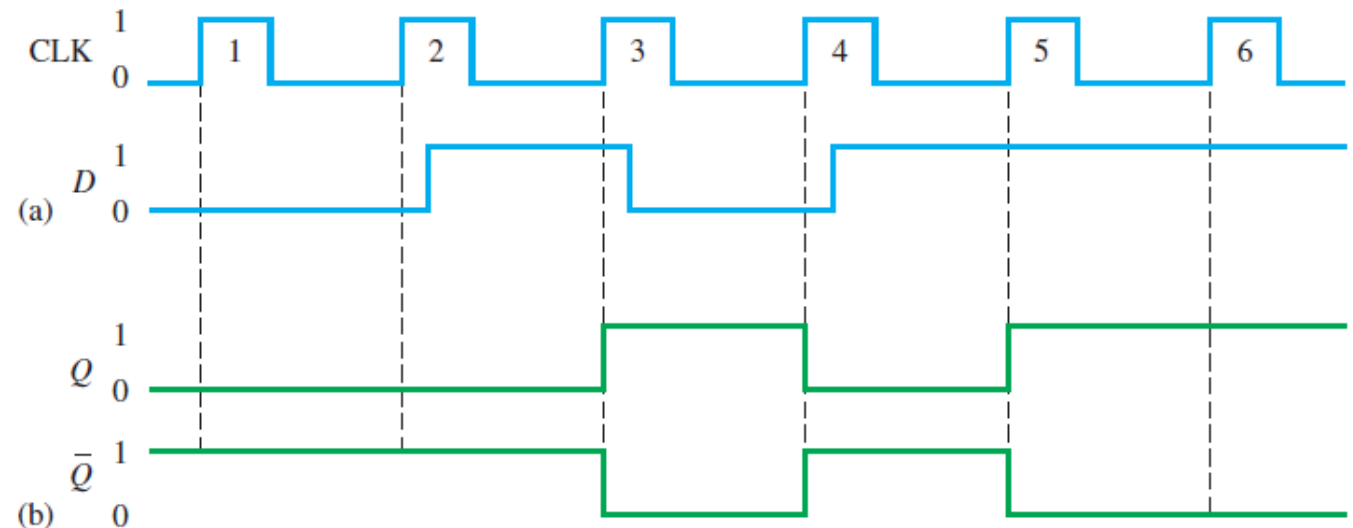
Outcome 1: To understand basics of Digital logic devices



Truth table for a positive edge-triggered D flip-flop.

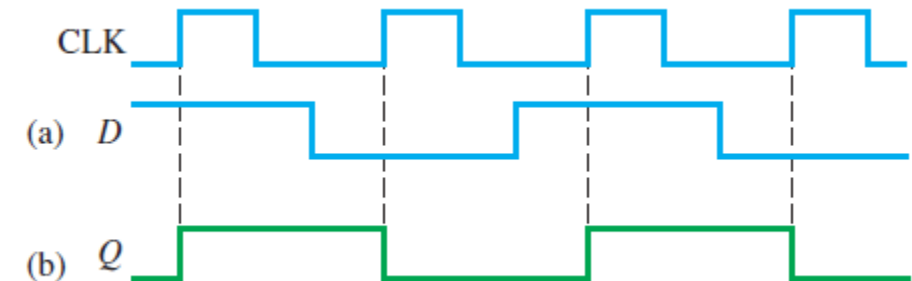
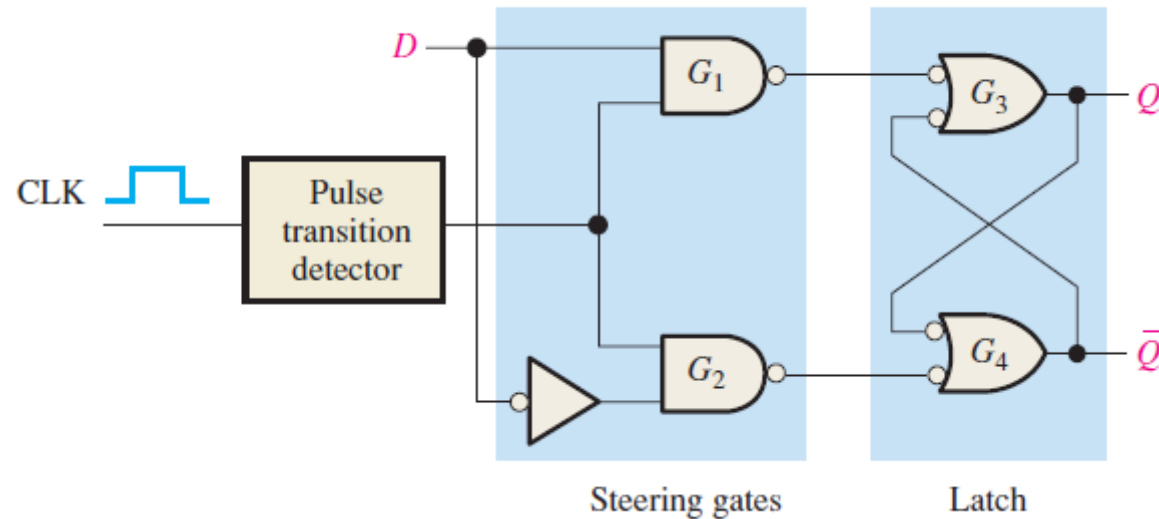
Inputs		Outputs		Comments
D	CLK	Q	$\bar{Q}$	
0	$\uparrow$	0	1	RESET
1	$\uparrow$	1	0	SET

$\uparrow$ = clock transition LOW to HIGH

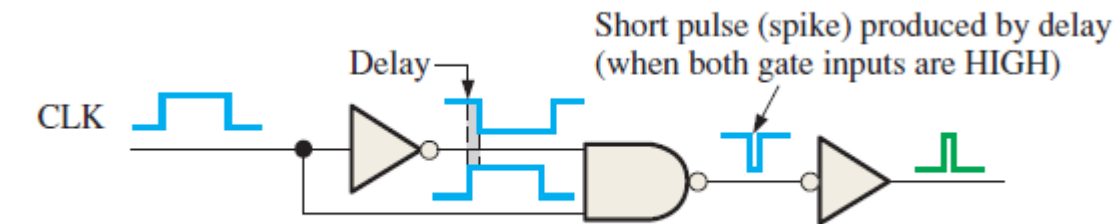


Outcome 1: To understand basics of Digital logic devices

74HC74



flip-flop starts out RESET.



(b) A type of pulse transition detector

Outcome 1: To understand basics of Digital logic devices

- J-K Flip-Flop

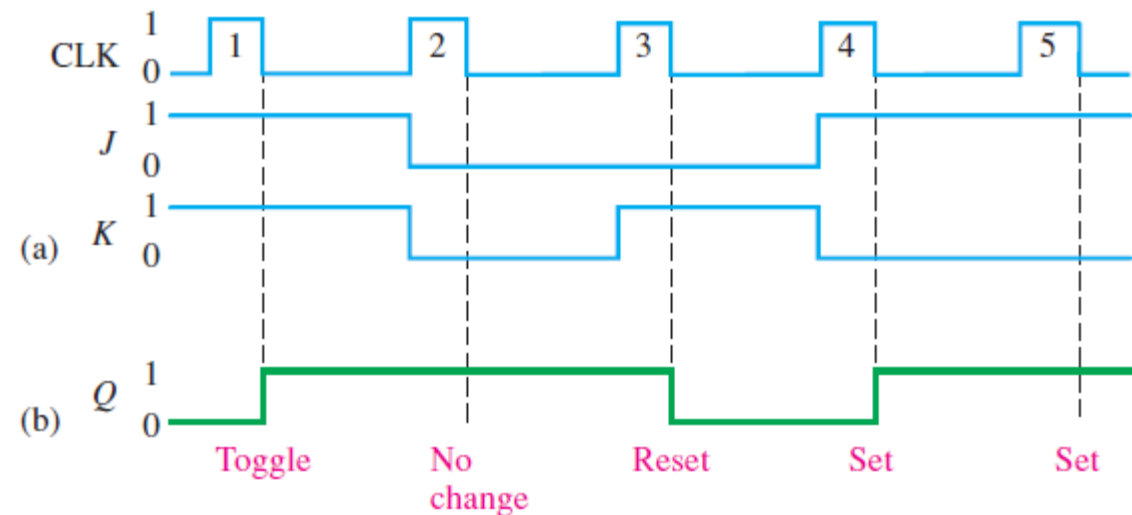
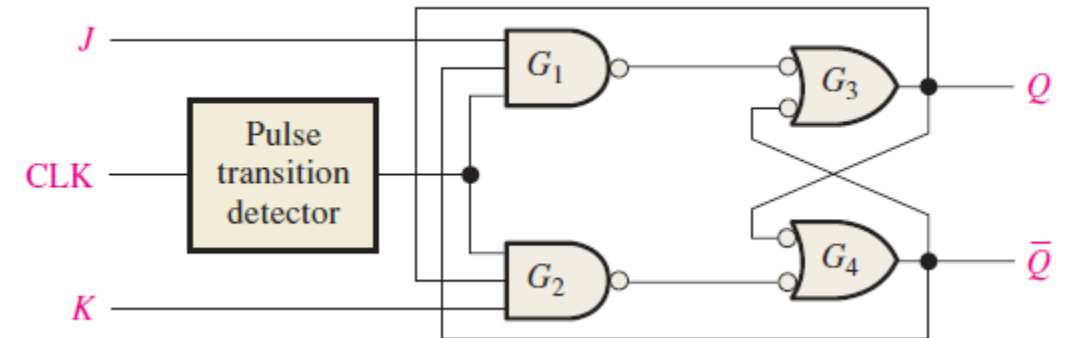
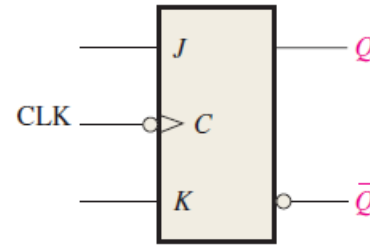
Truth table for a positive edge-triggered J-K flip-flop.

Inputs			Outputs		Comments
J	K	CLK	Q	$\bar{Q}$	
0	0	$\uparrow$	Q_0	$\bar{Q}_0$	No change
0	1	$\uparrow$	0	1	RESET
1	0	$\uparrow$	1	0	SET
1	1	$\uparrow$	$\bar{Q}_0$	Q_0	Toggle

$\uparrow$ = clock transition LOW to HIGH

Q_0 = output level prior to clock transition

A J-K flip-flop connected for toggle operation is sometimes called a *T flip-flop*.

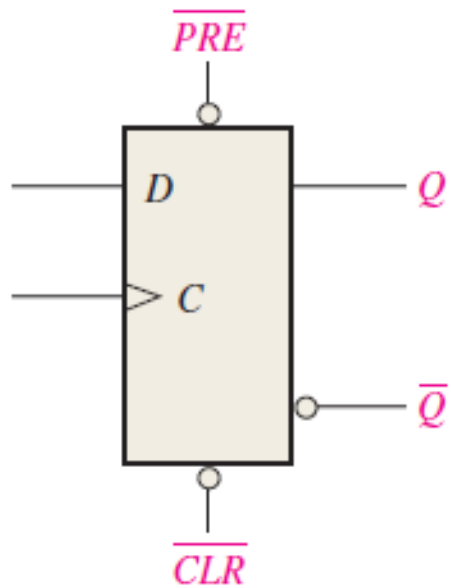


Outcome 1: To understand basics of Digital logic devices

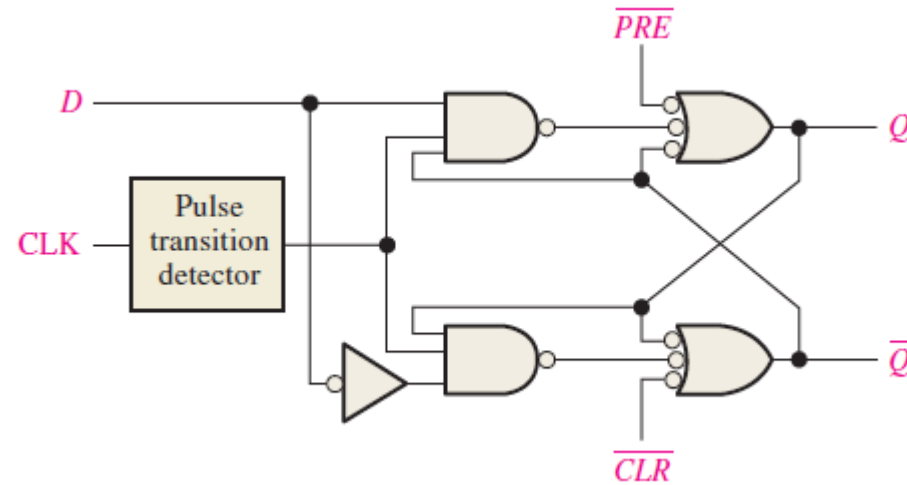
- Asynchronous Preset and Clear Inputs
- For the flip-flops, the D and J - K inputs are called *synchronous inputs* because data on these inputs are transferred to the flip-flop's output only on the triggering edge of the clock pulse; that is, the data are transferred synchronously with the clock.
- **asynchronous** inputs: inputs that affect the state of the flip-flop *independent of the clock*.
 - They are normally labeled **preset** (PRE) and **clear** (CLR), or *direct set* (SD) and *direct reset* (RD) by some manufacturers

Outcome 1: To understand basics of Digital logic devices

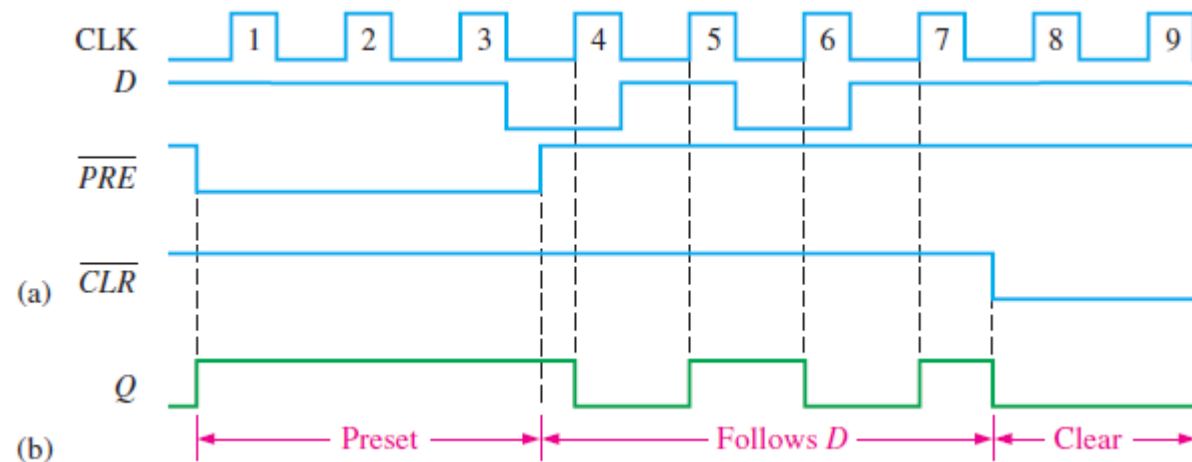
CMOS and bipolar (TTL) flip-flops.



Logic symbol for a D flip-flop with active-LOW preset and clear inputs.

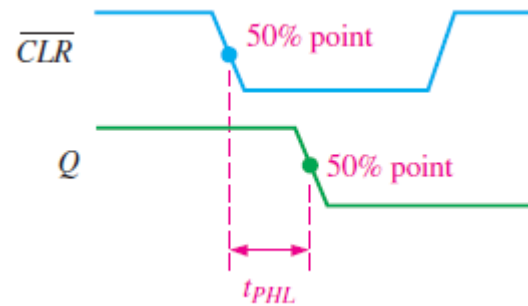
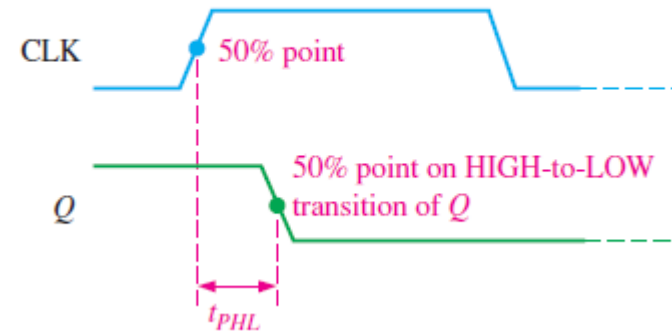
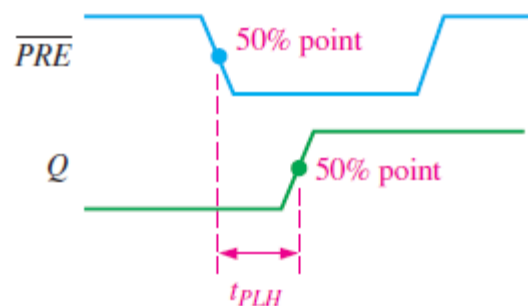
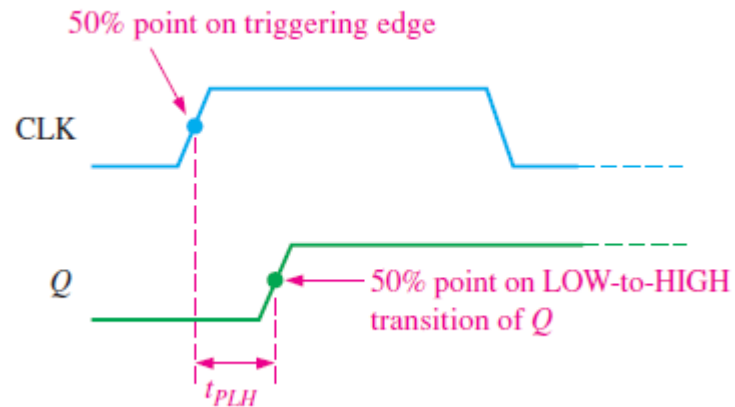


Logic diagram for a basic D flip-flop with active-LOW preset and clear inputs



Outcome 1: To understand basics of Digital logic devices

- Flip-Flop Operating Characteristics
- propagation delay time

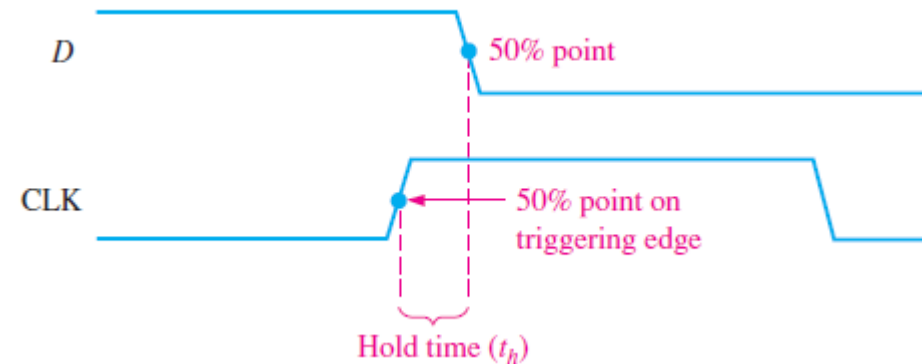
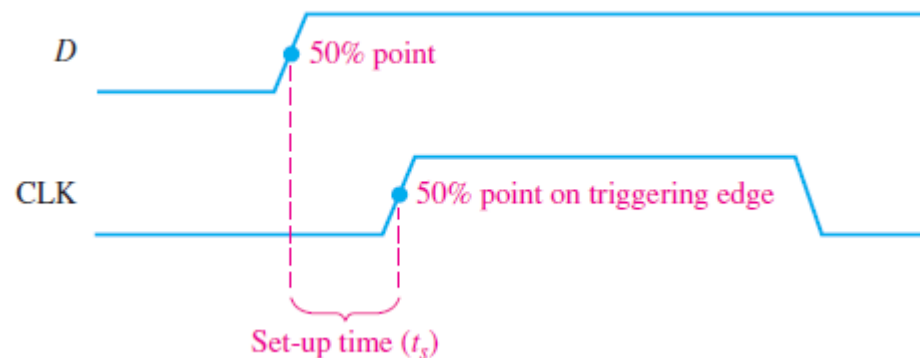


Propagation delays, clock to output.

Propagation delays, preset input to output and clear input to output.

Outcome 1: To understand basics of Digital logic devices

- Set-up Time and Hold Time
- The **set-up time** (t_s) and **hold time** (t_h) is the minimum interval required for the logic levels to be maintained constantly on the inputs (J and K , or D) prior and after, respectively, to the triggering edge of the clock pulse in order for the levels to be reliably clocked into the flip-flop.



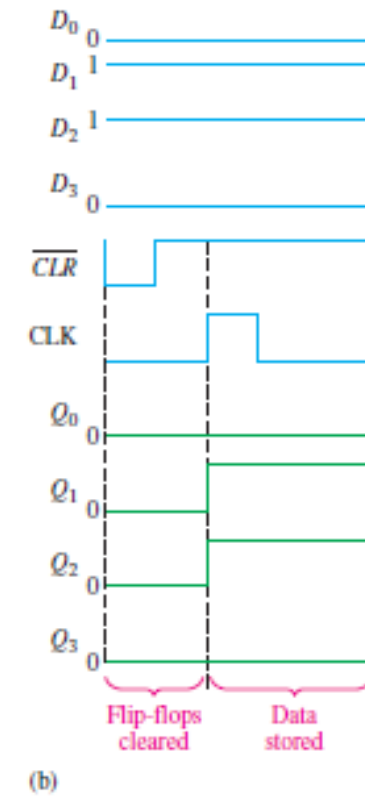
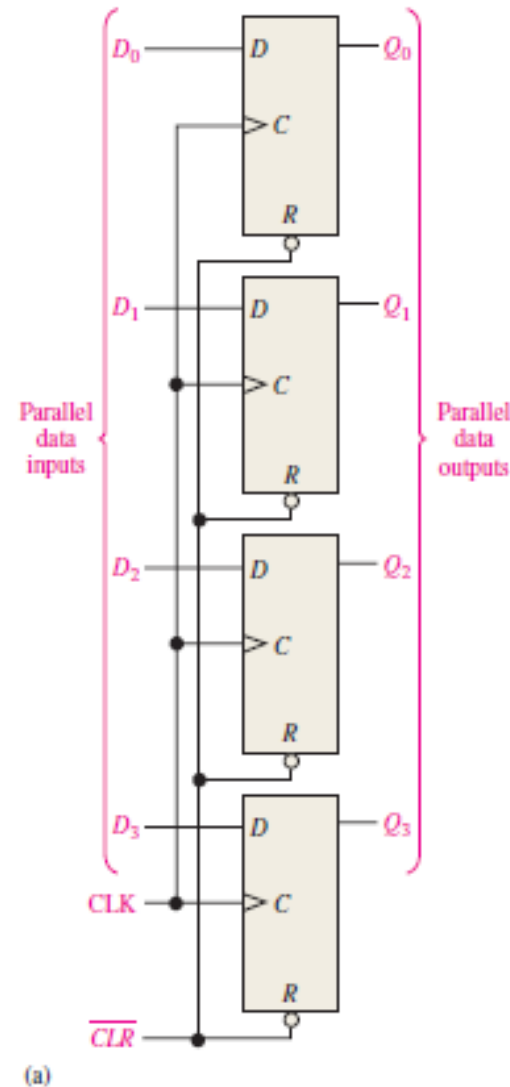
Outcome 1: To understand basics of Digital logic devices

- Maximum Clock Frequency
- The maximum clock frequency ($f_{\max}$) is the highest rate at which a flip-flop can be reliably triggered.
- Pulse Widths
- Minimum pulse widths (t_W) for reliable operation are usually specified by the manufacturer for the clock, preset, and clear inputs.
- Power Dissipation
- The **power dissipation** of any digital circuit is the total power consumption of the device.

$$P = V_{CC} \times I_{CC} = 5 \text{ V} \times 5 \text{ mA} = 25 \text{ mW} \quad P_T = 10 \times 25 \text{ mW} = 250 \text{ mW} = 0.25 \text{ W} \quad I = \frac{250 \text{ mW}}{5 \text{ V}} = 50 \text{ mA}$$

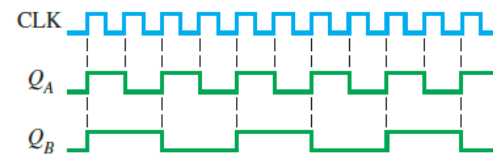
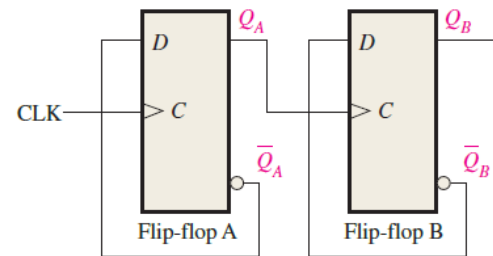
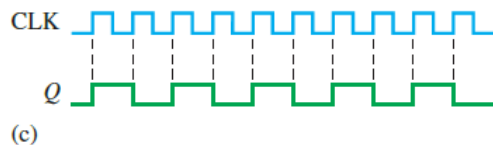
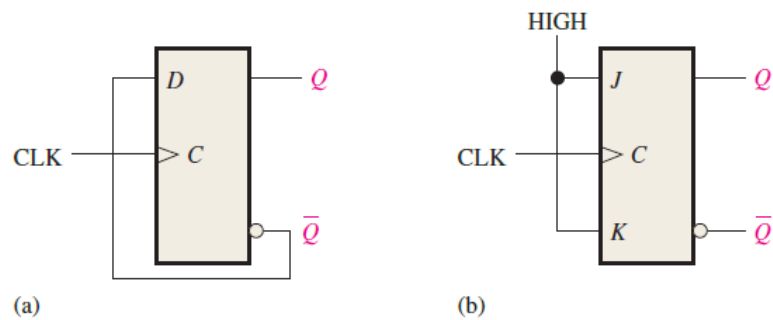
Outcome 2

- Flip-Flop Applications
- Parallel Data Storage



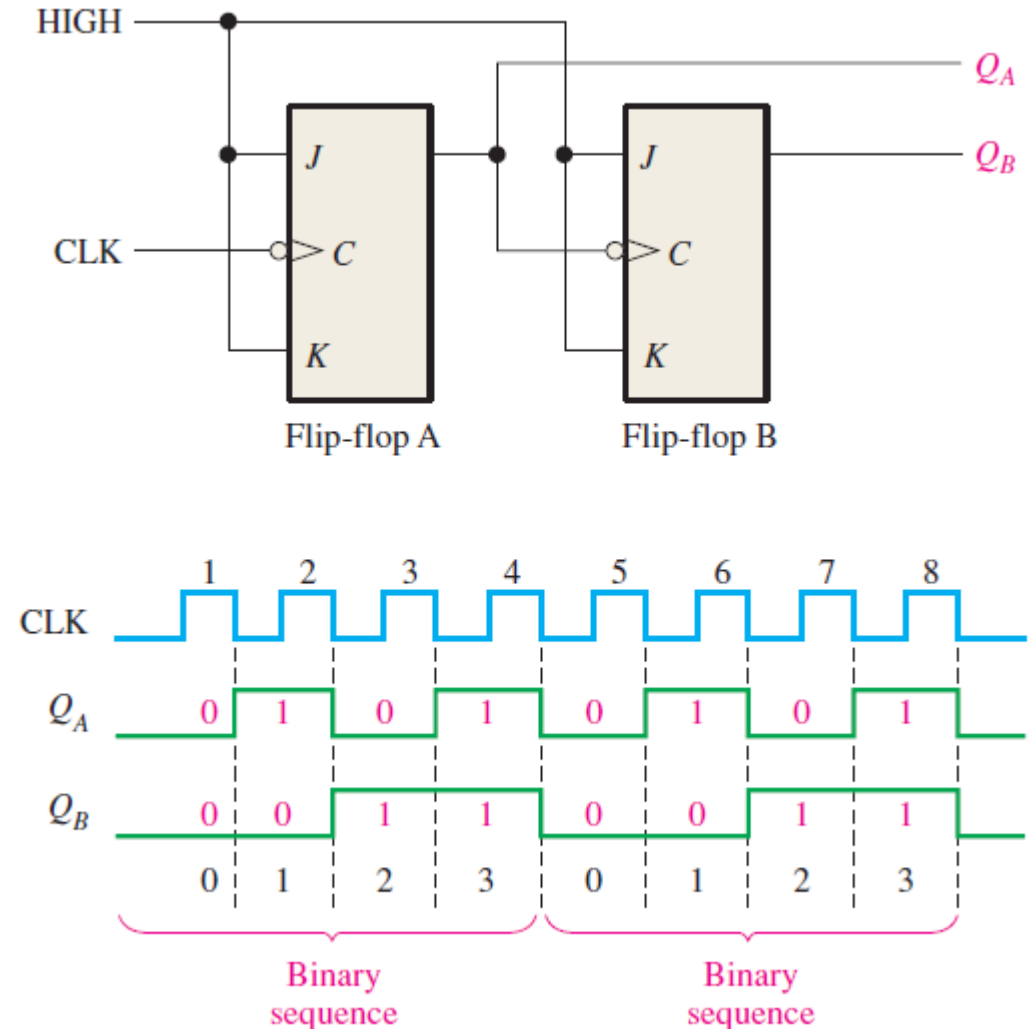
Outcome 2

- Frequency Division
- When a pulse waveform is applied to the clock input of a D or J-K flip-flop that is connected to toggle ($D = Q'$ or $J = K = 1$), the Q output is a square wave with one-half the frequency of the clock input.
 - a single flip-flop can be applied as a divide-by-2 device
 - a frequency division of $2n$ is achieved, where n is the number of flip-flops



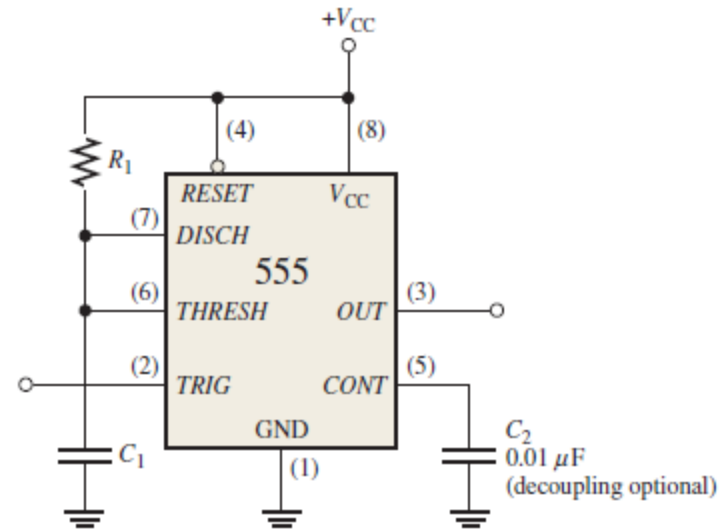
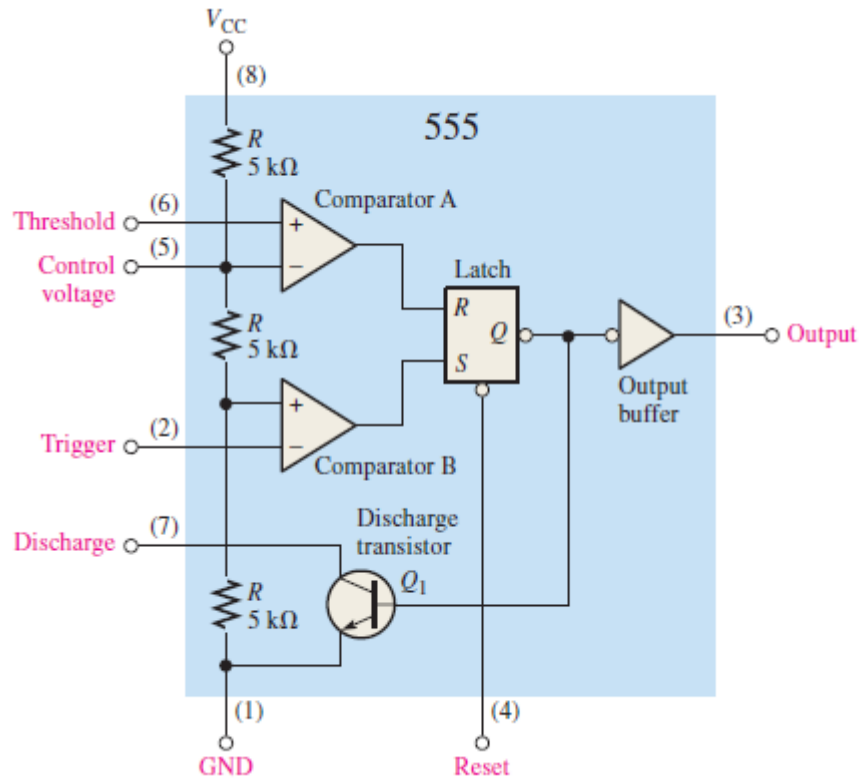
Outcome 2

- Counting
- Negative edge-triggered J-K flip-flops
- If we take Q_A as the least significant bit, a 2-bit sequence is produced as the flip-flops are clocked.
- This binary sequence repeats every four clock pulses



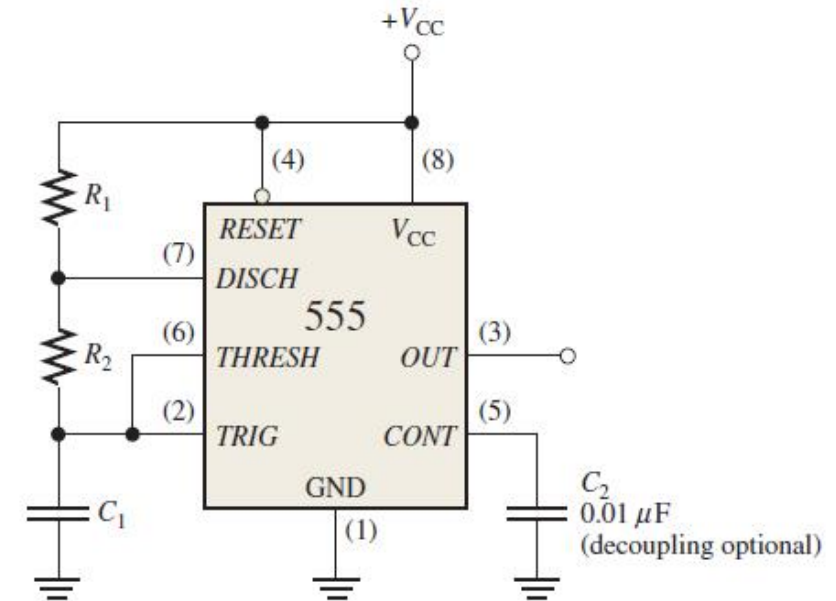
Outcome 2

Monostable (One-Shot) Operation



$$t_w = 1.1R_1C_1$$

Astable Operation



$$f = \frac{1.44}{(R_1 + 2R_2)C_1}$$